Maulana Abul Kalam Azad University of Technology, West Bengal
(Formerly West Bengal University of Technology)
B. Tech. in Computer Science and Business Systems
(Applicable from the Academic Session 2020-2021)

Curriculum Structure
Second Year Third Semester

Theory

Sl. No.	Type of course	Code	Course Title	Hours per week			Credits
				Lecture	Tutorial	Practical	
1	Engineering Science Course	ESC-301	Digital Electronics	3	0	0	3
2	Professional Core Courses	PCC-CSBS301	Data Structure & Algorithms	3	0	0	3
3	Professional Core Courses	PCC-CSBS302	Discrete Mathematics	3	1	0	4
4	Humanities & Social Sciences including Management courses	HSMC-301	Economics for Engineers	3	0	0	3
5	Basic Science course	BSC-301	Computational Statistics	3	0	0	3
				15	1	0	
						Total credits	**16**

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Practical

Sl. No.	Type of course	Code	Course Title	Hours per week			Credits
				Lecture	Tutorial	Practical	
1	Engineering Science Course	ESC-391	Digital Electronics Lab	0	0	2	1.5
2	Professional Core Courses	PCC-CSBS391	Data Structure & Algorithms Lab	0	0	4	3
3	Basic Science course	BSC-391	Computational Statistics Lab	0	0	2	1.5
				0	0	8	
			Total credits				**6**
			TOTAL CREDITS(Theory+Practical)				**22**

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Digital Electronics
Code: ESC-301
Contact: 3L

Name of the Course:	**Digital Electronics**		
Course Code: ESC-301	Semester: III		
Duration: 6 months	Maximum Marks: 100		
Teaching Scheme		**Examination Scheme**	
Theory: 3 hrs./week		Mid Semester exam: 15	
Tutorial: NIL		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam : 70 Marks	
Credit Points:	3		
Objective:			
1	To acquire the basic knowledge of different analog components and their applications		
2	To acquire the basic knowledge of digital logic levels and application of knowledge to understand digital electronics circuits.		
3	To prepare students to perform the analysis and design of various digital electronic circuits		
Pre-Requisite:			
1	Basic Electronics Parts I & II learned in the First year, semesters 1 & 2. Basic BJTs,.		
2	Basic concept of the working of P-N diodes, Schottky diodes,		
3	Basic FETs and OPAMP as a basic circuit component. Concept of Feedback		

Module	
1	a) Data and number systems; Binary, Octal and Hexadecimal representation and their conversions; BCD,ASCII, EBDIC, Gray codes and their conversions; Signed binary number representation with 1's and 2's complement methods, Binary arithmetic. [5] b) Venn diagram, Boolean algebra; Various Logic gates- their truth tables and circuits; Representation in SOP and POS forms; Minimization of logic expressions by algebraic method, K-map method [6]
2	a) Combinational circuits- Adder and Subtractor circuits; Applications and circuits of Encoder, Decoder, Comparator, Multiplexer, De-Multiplexer and Parity Generator. [5] b) Memory Systems: RAM, ROM, EPROM, EEROM [4] c) Design of combinational circuits-using ROM, Programming logic devices and gate arrays. (PLAs and PLDs) [4]

3	Sequential Circuits- Basic memory element-S-R, J-K, D and T Flip Flops, various types of Registers and counters and their design, Irregular counter, State table and state transition diagram, sequential circuits design methodology. [6]
4	a) Different types of A/D and D/A conversion techniques. [4] b) Logic families- TTL, ECL, MOS and CMOS, their operation and specifications. [6]

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DATA STRUCTURES AND ALGORITHMS (PCC-CSBS301)

Name of the Course:	**DATA STRUCTURES AND ALGORITHMS**	
Course Code: **PCC-CSBS301**	Semester: III	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: 2 hrs/week		End Semester Exam: 70 Marks
Credit Points:	4	
Objective:		
1		
2		
3		
4		
5		
6		
7		
Pre-Requisite:		
1		

Basic Terminologies and Introduction to Algorithm & Data Organisation: Algorithm specification, Recursion, Performance analysis, Asymptotic Notation - The Big-O, Omega and Theta notation, Programming Style, Refinement of Coding - Time-Space Trade Off, Testing, Data Abstraction

Linear Data Structure: Array, Stack, Queue, Linked-list and its types, Various Representations, Operations & Applications of Linear Data Structures

Non-linear Data Structure: Trees (Binary Tree, Threaded Binary Tree, Binary Search Tree, B & B+ Tree, AVL Tree, Splay Tree) and Graphs (Directed, Undirected), Various Representations, Operations & Applications of Non-Linear Data Structures

Searching and Sorting on Various Data Structures: Sequential Search, Binary Search, Comparison Trees, Breadth First Search, Depth First Search Insertion Sort, Selection

Sort, Shell Sort, Divide and Conquer Sort, Merge Sort, Quick Sort, Heapsort, Introduction to Hashing

File: Organisation (Sequential, Direct, Indexed Sequential, Hashed) and various types of accessing schemes.

Graph: Basic Terminologies and Representations, Graph search and traversal algorithms and complexity analysis.

Text Books:

1. Fundamentals of Data Structures, E. Horowitz, S. Sahni, S. A-Freed, Universities Press.
2. Data Structures, R.S. Salaria, Khanna Book Publishing, Delhi.
3. Data Structures and Algorithms, A. V. Aho, J. E. Hopperoft, J. D. UIlman, Pearson.
4. Expert Data Structures with C, R.P. Patel, Khanna Publishing House.

Reference Books:

1. The Art of Computer Programming: Volume 1: Fundamental Algorithms, Donald E. Knuth.
2. Design and Analysis of Algorithms, Gajendra Sharma, Khanna Book Publishing
3. Introduction to Algorithms, Thomas, H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, The MIT Press.
4. Open Data Structures: An Introduction (Open Paths to Enriched Learning), (Thirty First Edition), Pat Morin, UBC Press.

Discrete Mathematics (PCC-CSBS302)

Boolean algebra: Introduction of Boolean algebra, truth table, basic logic gate, basic postulates of Boolean algebra, principle of duality, canonical form, Karnaugh map.

Abstract algebra: Set, relation, group, ring, field.

Combinatorics: Basic counting, balls and bins problems, generating functions, recurrence relations. Proof techniques, principle of mathematical induction, pigeonhole principle.

Graph Theory: Graphs and digraphs, complement, isomorphism, connectedness and reachability, adjacency matrix, Eulerian paths and circuits in graphs and digraphs, Hamiltonian paths and circuits in graphs and tournaments, trees; Planar graphs, Euler's formula, dual of a planer graph, independence number and clique number, chromatic number, statement of Four-color theorem.

Logic: Propositional calculus - propositions and connectives, syntax; Semantics - truth assignments and truth tables, validity and satisfiability, tautology; Adequate set of connectives; Equivalence and normal forms; Compactness and resolution; Formal reducibility - natural deduction system and axiom system; Soundness and completeness.

Text Books:

1. Topics in Algebra, I. N. Herstein, John Wiley and Sons.
2. Digital Logic & Computer Design, M. Morris Mano, Pearson.
3. Elements of Discrete Mathematics, (Second Edition) C. L. LiuMcGraw Hill, New Delhi.
4. Graph Theory with Applications, J. A. Bondy and U. S. R. Murty, Macmillan Press, London.
5. Mathematical Logic for Computer Science, L. Zhongwan, World Scientific, Singapore.
6. Discrete Structures, S.B. Singh, Khanna Publishing House.

Reference Books:

1. Introduction to linear algebra. Gilbert Strang.
2. Introductory Combinatorics, R. A. Brualdi, North-Holland, New York.
3. Graph Theory with Applications to Engineering and Computer Science, N. Deo, Prentice Hall, Englewood Cliffs.
4. Introduction to Mathematical Logic, (Second Edition), E. Mendelsohn, Van-Nostrand, London.

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Economics for Engineers (Humanities-II) Code: HSMC-301
Contacts: 3L

Name of the Course:	**Economics for Engineers (Humanities-II)**		
Course Code: HSMC-301	Semester: III		
Duration: 6 months	Maximum Marks: 100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: NIL		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: NIL		End Semester Exam: 70 Marks	
Credit Points:	3		

Objective:	
1	Understand the role and scope of Engineering Economics and the process of economic decision making
2	Understand the different concepts of cost and different cost estimation techniques
3	Familiarization with the concepts of cash flow, time value of money and different interest formulas
4	Appreciation of the role of uncertainty in future events and using different concepts from probability to deal with uncertainty
5	Understand the concepts of Depreciation and Replacement analysis along with their methods of calculation
6	Familiarization with the phenomenon of inflation and the use of price indices in engineering Economics
7	Introduction to basic concepts of Accounting and Financial Management
Pre-Requisite:	
1	Mathematics

Unit	Content	Hrs/Unit	Marks/Unit

1	1. Economic Decisions Making – Overview, Problems, Role, Decision making process. 2. Engineering Costs & Estimation – Fixed, Variable, Marginal & Average Costs, Sunk Costs, Opportunity Costs, Recurring And Nonrecurring Costs, Incremental Costs, Cash Costs vs Book Costs, Life-Cycle Costs; Types Of Estimate, Estimating Models - Per- Unit Model, Segmenting Model, Cost Indexes, Power-Sizing Model, Improvement & Learning Curve, Benefits.	9	
2	3. Cash Flow, Interest and Equivalence: Cash Flow – Diagrams, Categories & Computation, Time Value of	9	

COMPUTATIONAL STATISTICS (BSC-301)

Name of the Course:	**COMPUTATIONAL STATISTICS**		
Course Code: **(BSC-301)**	Semester: III		
Duration: 6 months	Maximum Marks: 100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: NIL		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
		End Semester Exam: 70 Marks	
Credit Points:	4		
Objective:			
1			
2			
3			
4			
5			
6			
7			
Pre-Requisite:			
1			

Multivariate Normal Distribution: Multivariate Normal Distribution Functions, Conditional Distribution and its relation to regression model, Estimation of parameters.

Multiple Linear Regression Model: Standard multiple regression models with emphasis on detection of collinearity, outliers, non-normality and autocorrelation, Validation of model assumptions.

Multivariate Regression: Assumptions of Multivariate Regression Models, Parameter estimation, Multivariate Analysis of variance and covariance

Discriminant Analysis: Statistical background, linear discriminant function analysis, Estimating linear discriminant functions and their properties.

Principal Component Analysis: Principal components, Algorithm for conducting principal component analysis, deciding on how many principal components to retain, H-plot.

Factor Analysis: Factor analysis model, Extracting common factors, determining number of factors, Transformation of factor analysis solutions, Factor scores.

Cluster Analysis: Introduction, Types of clustering, Correlations and distances, clustering by partitioning methods, hierarchical clustering, overlapping clustering, K-Means Clustering-Profiling and Interpreting Clusters.

Text Books:

1. An Introduction to Multivariate Statistical Analysis, T.W. Anderson.
2. Applied Multivariate Data Analysis, Vol I & II, J.D. Jobson.
3. Statistical Tests for Multivariate Analysis, H. Kris.
4. Programming Python, Mark Lutz.
5. Python 3 for Absolute Beginners, Tim Hall and J-P Stacey.
6. Beginning Python: From Novice to Professional, Magnus Lie Hetland. Edition, 2005.
7. Beginner's Guide for Data Analysis using R Programming, Jeeva Jose, Khanna Publishing House.
8. Data Science and Analytics, V.K. Jain, Khanna Publishing House.

Reference Books:

1. Regression Diagnostics , Identifying Influential Data and Sources of Collinearety, D.A. Belsey, E. Kuh and R.E. Welsch
2. Applied Linear Regression Models, J. Neter, W. Wasserman and M.H. Kutner.
3. The Foundations of Factor Analysis, A.S. Mulaik.
4. Introduction to Linear Regression Analysis, D.C. Montgomery and E.A. Peck.
5. Cluster Analysis for Applications, M.R. Anderberg.
6. Multivariate Statistical Analysis, D.F. Morrison.
7. Python for Data Analysis, Wes Mc Kinney.

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Name of the Course:	**COMPUTATIONAL STATISTICS**	
Course Code: **(BSC-391)**	Semester: III	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme	**Examination Scheme**	
Practical: 3 hrs/week	Continuous Internal Assessment: 40	
	External Assesement: 60	
Credit Points:	1.5	
Objective:		
1		
2		
3		
4		
5		
6		
7		
Pre-Requisite:		
1		

Laboratory

Python Concepts, Data Structures, Classes: Interpreter, Program Execution, Statements, Expressions, Flow Controls, Functions, Numeric Types, Sequences and Class Definition, Constructors, Text & Binary Files - Reading and Writing

Visualization in Python: Matplotlib package, Plotting Graphs, Controlling Graph, Adding Text, More Graph Types, Getting and setting values, Patches

Multivariate data analysis: Multiple regression, multivariate regression, cluster analysis with various algorithms, factor analysis, PCA and linear discriminant analysis. Various datasets should be used for each topic

Name of the Course:	**DATA STRUCTURES AND ALGORITHMS Lab**	
Course Code: **PCC-CSBS391**	Semester: III	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Practical: 4 hrs/week		Continuous Internal Assessment: 40
		External Assesement: 60
Credit Points:	2	

Objective:	
1	
2	
3	
4	
5	
6	
7	

Pre-Requisite:	
1	**PCC-CSBS301**

Laboratory Experiments:	
1	Towers of Hanoi using user defined stacks.
2	Reading, writing, and addition of polynomials.
3	Line editors with line count, word count showing on the screen.

4	Trees with all operations.
5	All graph algorithms.
6	Saving / retrieving non-linear data structure in/from a file

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Digital Electronics Lab (ESC-391)

Name of the Course:	**Digital Electronics Lab**	
Course Code: ESC-391	Semester: III	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Practical: 3 hrs/week		Continuous Internal Assessment: 40
		External Assesement: 60
Credit Points:	1.5	

Objective:	
1	
2	
3	
4	
5	

Pre-Requisite:	
1	ESC-301

1	Realization of basic gates using Universal logic gates.
2	Code conversion circuits- BCD to Excess-3 and vice-versa.
3	Four-bit parity generator and comparator circuits.
4	Construction of simple Decoder and Multiplexer circuits using logic gates.
5	Design of combinational circuit for BCD to decimal conversion to drive 7-segment display using multiplexer.

6	Construction of simple arithmetic circuits-Adder, Subtractor.
7	Realization of RS-JK and D flip-flops using Universal logic gates.
8	Realization of Universal Register using JK flip-flops and logic gates.
9	Realization of Universal Register using multiplexer and flip-flops.
10	Construction of Adder circuit using Shift Register and full Adder.
11	Realization of Asynchronous Up/Down counter.
12	Realization of Synchronous Up/Down counter.
13	Design of Sequential Counter with irregular sequences.
14	Realization of Ring counter and Johnson's counter.
15	Construction of adder circuit using Shift Register and full Adder.

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Curriculum Structure
Second Year Fourth Semester

Theory

Sl. No.	Type of course	Code	Course Title	Hours per week			Credits
				L	T	P	
1		HSMC-401	Introduction to Innovation, IP Management & Entrepreneurship	2	1	0	3
2	Professional Core Courses	PCC-CSBS401	FORMAL LANGUAGE & AUTOMATA THEORY	2	1	0	3
3	Professional Core Courses	PCC-CSBS402	OBJECT ORIENTED PROGRAMMING	2	1	0	3
4	Professional Core Courses	PCC-CSBS403	COMPUTER ORGANIZATION & ARCHITECHTURE	2	1	0	3
5	Basic Science course	*BSC 401*	Biology	2	1	0	3
6	Mandatory Course	MC 401	Environmental Science	1	0	0	1
							16

Practical

Sl. No.	Type of course	Code	Course Title	Hours per week			Credits
				Lecture	Tutorial	Practical	
1		ESC-481	DESIGN THINKING	0	1	3	2.5
2		PCC -492	Object Oriented Programming with C++ Lab	0	0	4	2
3		PCC- CSBS 493	COMPUTER ORGANIZATION & ARCHITECHTURE Lab	0	0	3	1.5
			Total credits				**6**
			TOTAL CREDITS(Theory+Practical)				**22**

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Introduction to Innovation, IP Management & Entrepreneurship (HSMC-401)

Course Pre Requisite(s):
Good knowledge of Fundamentals of Management

Course Outcome(s):
The major emphasis of the course will be on creating a learning system through which management students can enhance their innovation and creative thinking skills, acquaint themselves with the special challenges of starting new ventures and use IPR as an effective tool to protect their innovations and intangible assets from exploitation.
As a part of this course, students will:

- Learn to be familiar with creative and innovative thinking styles
- Learn to investigate, understand and internalize the process of founding a startup
- Learn to manage various types of IPR to protect competitive advantage

Topics to Be Covered:

UNIT – I
Innovation: What and Why?
Innovation as a core business process, Sources of innovation, Knowledge push vs. need pull innovations.

Class Discussion- Is innovation manageable or just a random gambling activity?

UNIT – II
Building an Innovative Organization
Creating new products and services, Exploiting open innovation and collaboration, Use of innovation for starting a new venture.
Class Discussion- Innovation: Co-operating across networks vs. 'go-it-alone' approach

UNIT – III
Entrepreneurship:

- Opportunity recognition and entry strategies
- Entrepreneurship as a Style of Management
- Maintaining Competitive Advantage- Use of IPR to protect Innovation

UNIT – IV
Entrepreneurship- Financial Planning:

- Financial Projections and Valuation
- Stages of financing
- Debt, Venture Capital and other forms of Financing

UNIT – V
Intellectual Property Rights (IPR)

- Introduction and the economics behind development of IPR: Business Perspective
- IPR in India – Genesis and Development

- International Context
- Concept of IP Management, Use in marketing

UNIT – VI
Types of Intellectual Property

- Patent- Procedure, Licensing and Assignment, Infringement and Penalty
- Trademark- Use in marketing, example of trademarks- Domain name
- Geographical Indications- What is GI, Why protect them?
- Copyright- What is copyright
- Industrial Designs- What is design? How to protect?

Class Discussion- Major Court battles regarding violation of patents between corporate companies

Home Assignment:
Case study materials book will be given to students. Students are required to meet in groups before coming to class and prepare on the case for the day. Instructor may ask the student groups to present their analysis and findings to the class.

Further, the topic for class discussion will be mentioned beforehand and students should be ready to discuss these topics (in groups) in class. Students are required to meet in groups before coming to class and prepare on the topic. Few topics are mentioned below as examples. Instructor can add or change any topic as per requirement.

Topic 1- Is innovation manageable or just a random gambling activity?
Topic 2- Innovation: Co-operating across networks vs. 'go-it-alone' approach
Topic 3- Major Court battles regarding violation of patents between corporate companies

Text Books:
1. Joe Tidd, John Bessant. Managing Innovation: Integrating Technological, Market and Organizational Change
2. Case Study Materials: To be distributed for class discussion

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FORMAL LANGUAGE & AUTOMATA THEORY (PCC-CSBS401)

Introduction: Alphabet, languages and grammars, productions and derivation, Chomsky hierarchy of languages.

Regular languages and finite automata: Regular expressions and languages, deterministic finite automata (DFA) and equivalence with regular expressions, nondeterministic finite automata (NFA) and equivalence with DFA, regular grammars and equivalence with finite automata, properties of regular languages, Kleene's theorem, pumping lemma for regular languages, Myhill-Nerode theorem and its uses, minimization of finite automata.

Context-free languages and pushdown automata: Context-free grammars (CFG) and languages (CFL), Chomsky and Greibach normal forms, nondeterministic pushdown automata (PDA) and equivalence with CFG, parse trees, ambiguity in CFG, pumping lemma for context-free languages, deterministic pushdown automata, closure properties of CFLs.

Context-sensitive languages: Context-sensitive grammars (CSG) and languages, linear bounded automata and equivalence with CSG.

Turing machines: The basic model for Turing machines (TM), Turing recognizable (recursively enumerable) and Turing-decidable (recursive) languages and their closure properties, variants of Turing machines, nondeterministic TMs and equivalence with deterministic TMs, unrestricted grammars and equivalence with Turing machines, TMs as enumerators.

Undecidability: Church-Turing thesis, universal Turing machine, the universal and diagonalization languages, reduction between languages and Rice s theorem, undecidable problems about languages.

Basic Introduction to Complexity: Introductory ideas on Time complexity of deterministic and nondeterministic Turing machines, P and NP, NP- completeness, Cook's Theorem, other NP - Complete problems.

Text Books:

1. Introduction to Automata Theory, Languages, and Computation John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman.

Reference Books:

1. Elements of the Theory of Computation, Harry R. Lewis and Christos H. Papadimitriou.
2. Automata and Computability, Dexter C. Kozen.
3. Introduction to the Theory of Computation, Michael Sipser.
4. Introduction to Languages and the Theory of Computation, John Martin.
5. Computers and Intractability: A Guide to the Theory of NP Completeness, M. R. Garey and D. S. Johnson.

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OBJECT ORIENTED PROGRAMMING (PCC-CSBS402)

Procedural programming, An Overview of C: Types Operator and Expressions, Scope and Lifetime, Constants, Pointers, Arrays, and References, Control Flow, Functions and Program Structure, Namespaces, error handling, Input and Output (C-way), Library Functions (string, math, stdlib), Command line arguments, Pre-processor directive

Some difference between C and C++: Single line comments, Local variable declaration within function scope, function declaration, function overloading, stronger type checking, Reference variable, parameter passing – value vs. reference, passing pointer by value or reference, #define constant vs const, Operator new and delete, the typecasting operator, Inline Functions in contrast to macro, default arguments

The Fundamentals of Object Oriented Programming: Necessity for OOP, Data Hiding, Data Abstraction, Encapsulation, Procedural Abstraction, Class and Object.

More extensions to C in C++ to provide OOP Facilities: Scope of Class and Scope Resolution Operator, Member Function of a Class, private, protected and public Access Specifier, this Keyword, Constructors and Destructors, friend class, error handling (exception)

Essentials of Object Oriented Programming: Operator overloading, Inheritance – Single and Multiple, Class Hierarchy, Pointers to Objects, Assignment of an Object to another Object, Polymorphism through dynamic binding, Virtual Functions, Overloading, overriding and hiding, Error Handling

Generic Programming: Template concept, class template, function template, template specialization

Input and Output: Streams, Files, Library functions, formatted output

Object Oriented Design and Modeling: UML concept, Use case for requirement capturing, Class diagram, Activity diagram and Sequence Diagram for design, Corresponding C++ code from design

Text Books:
1. The C++ Programming Language, Bjarne Stroustrup, Addison Wesley.
2. C++ and Object-Oriented Programming Paradigm, Debasish Jana, PHI Learning Pvt. Ltd.

Reference Books:
1. Programming – Principles and Practice Using C++, Bjarne Stroustrup, Addison Wesley.
2. The Design and Evolution of C++, Bjarne Stroustrup, Addison Wesley.

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COMPUTER ORGANIZATION & ARCHITECHTURE (PCC-CSBS403)

Revision of basics in Boolean logic and Combinational/Sequential Circuits.

Functional blocks of a computer: CPU, memory, input-output subsystems, control unit.

Instruction set architecture of a CPU: Registers, instruction execution cycle, RTL interpretation of instructions, addressing modes, instruction set. Outlining instruction sets of some common CPUs.

Data representation: Signed number representation, fixed and floating point representations, character representation.

Computer arithmetic: Integer addition and subtraction, ripple carry adder, carry look-ahead adder, etc. multiplication – shift-and-add, Booth multiplier, carry save multiplier, etc. Division restoring and non-restoring techniques, floating point arithmetic, IEEE 754 format.

Introduction to x86 architecture.

CPU control unit design: Hardwired and micro-programmed design approaches, design of a simple hypothetical CPU.

Memory system design: Semiconductor memory technologies, memory organization.

Peripheral devices and their characteristics: Input-output subsystems, I/O device interface, I/O transfers – program controlled, interrupt driven and DMA, privileged and non-privileged instructions, software interrupts and exceptions. Programs and processes – role of interrupts in process state transitions, I/O device interfaces – SCII, USB

Pipelining: Basic concepts of pipelining, throughput and speedup, pipeline hazards.
Parallel Processors: Introduction to parallel processors, Concurrent access to memory and cache coherency.

Memory organization: Memory interleaving, concept of hierarchical memory organization, cache memory, cache size vs. block size, mapping functions, replacement algorithms, write policies.

Text Books:

1. Computer System Architecture M. M. Mano:, 3rd ed., Prentice Hall of India, New Delhi, 1993.
2. Computer Organization and Design: The Hardware/Software Interface, David A. Patterson and John L. Hennessy.
3. Computer Organization and Embedded Systems, Carl Hamacher.

Reference Books:

1. Computer Architecture and Organization, John P. Hayes.
2. Computer Organization and Architecture: Designing for Performance, William Stallings.
3. Computer System Design and Architecture, Vincent P. Heuring and Harry F. Jordan.

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Biology
Code: BSC 401
Contacts: 2L+1T

Name of the Course:	***Biology***	
Course Code: BSC-401	*Semester: IV*	
Duration: 6 months	*Maximum Marks:100*	
Teaching Scheme		***Examination Scheme***
Theory: 2hrs./week		*Mid Semester exam: 15*
Tutorial: 1 hour		*Assignment and Quiz: 10 marks*
		Attendance: 5 marks
Practical: NIL		*End Semester Exam: 70 Marks*
Credit Points:	*3*	
Objective:		
1	*Bring out the fundamental differences between science and engineering*	
2	*Discuss how biological observations of 18_{th} Century that lead to major discoveries*	
Pre-Requisite:		
1	*Basic knowledge of Physics ,Chemistry and mathematics*	

Unit	*Content*	*Hrs/Unit*	*Marks/Unit*
1	*To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18_{th} Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.*	*2*	

2	*The underlying criterion, such as morphological, biochemical or ecological be highlighted. Hierarchy of life forms at phenomenological level. A common thread weaves this hierarchy Classification. Discuss classification based on (a) cellularity- Unicellular or multicellular (b) ultrastructure- prokaryotes or eucaryotes. (c)*	*3*	
	energy and Carbon utilisation -Autotrophs, heterotrophs, lithotropes (d) Ammonia excretion – aminotelic, uricoteliec, ureotelic (e) Habitata- acquatic or terrestrial (e) Molecular taxonomy- three major kingdoms of life. A given organism can come under different category based on classification. Model organisms for the study of biology come from different groups. E.coli, S.cerevisiae, D. Melanogaster, C. elegance, A. Thaliana, M. musculus		
3	*To convey that "Genetics is to biology what Newton'slawsareto Physical Sciences" Mendel's laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Gene interaction, Epistasis. Meiosis and Mitosis be taught as a part of genetics. Emphasis to be give not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Concepts of recessiveness and dominance. Concept of mapping of phenotype to genes. Discuss about the single gene disorders in humans. Discuss the concept of complementation using human genetics.*	***4***	
4.	*Biomolecules: To convey that all forms of life have the same building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. In this context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA.Two carbon units and lipids.*	***4***	

5	*Enzymes: To convey that without catalysis life would not have existed on earth* *Enzymology: How to monitor enzyme catalysed reactions. How does an enzyme catalyse reactions? Enzyme classification. Mechanism of enzyme action. Discuss at least two examples. Enzyme kinetics and kinetic parameters.* *Why should we know these parameters to understand biology? RNA catalysis.*	***4***	
6	*Information Transfer:The molecular basis of coding and decoding genetic information is universal* *Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA*	***4***	
	structure- from single stranded to double helix to nucleosomes. Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination.		
7	*Macromolecular analysis: How to analyse biological processes at the reductionist level* *Proteins- structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Proteins as enzymes, transporters, receptors and structural elements.*	***5***	
8	*Metabolism: The fundamental principles of energy transactions are the same in physical and biological world.* *Thermodynamics as applied to biological systems. Exothermic and endothermic versus endergonic and exergoinc reactions. Concept of Keqand its relation to standard free energy. Spontaneity. ATP as an energy currency. This should include the breakdown of glucose to CO2 + H2O (Glycolysis and Krebs cycle) and synthesis of glucose from CO2 and H2O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge*	***4***	

9	*Microbiology Concept of single celled organisms. Concept of species and strains. Identification and classification of microorganisms. Microscopy. Ecological aspects of single celled organisms. Sterilization and media compositions. Growth kinetics.*	***3***	

Text books/ reference books:

1. Biology: A global approach: Campbell, N. A.; Reece, J. B.; Urry, Lisa; Cain, M, L.; Wasserman, S. A.; Minorsky, P. V.; Jackson, R. B. Pearson Education Ltd
2. Outlines of Biochemistry, Conn, E.E; Stumpf, P.K; Bruening, G; Doi, R.H. John Wiley and Sons
3. Principles of Biochemistry (V Edition), By Nelson, D. L.; and Cox, M. M.W.H. Freeman and Company
4. Molecular Genetics (Second edition), Stent, G. S.; and Calender, R. W.H. Freeman and company, Distributed by Satish Kumar Jain for CBS Publisher
5. Microbiology, Prescott, L.M J.P. Harley and C.A. Klein 1995. 2nd edition Wm, C. Brown Publishers

Course Outcomes:

On completion of the course students will be able to

BSC-401.1 Describe how biological observations of 18_{th} Century that lead to major discoveries.

BSC-401.2 Convey that classification per se is not what biology is all about but highlight the underlying criteria, such as morphological, biochemical and ecological

BSC-401.3 Highlight the concepts of recessiveness and dominance during the passage of genetic material from parent to offspring

BSC-401.4 Convey that all forms of life have the same building blocks and yet the manifestations are as diverse as one can imagine

BSC-401.5 Classify enzymes and distinguish between different mechanisms of enzyme action.

BSC-401.6 Identify DNA as a genetic material in the molecular basis of information transfer.

BSC-401.7 Analyse biological processes at the reductionistic level

BSC-401.8 Apply thermodynamic principles to biological systems.

BSC-401.9 Identify and classify microorganisms.

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Environmental Sciences
Code: MC-401
Contacts: 1L

Name of the Course:	***Environmental Sciences***		
Course Code: MC-401	*Semester: IV*		
Duration:6 months	*Maximum Marks:100*		
Teaching Scheme		***Examination Scheme***	
Theory:1hrs./week		*Mid Semester exam: 15*	
Tutorial: NIL		*Assignment and Quiz : 10 marks*	
		Attendance : 5 marks	
Practical: NIL		*End Semester Exam :70 Marks*	
Credit Points:	*1*		
Objective:			
1	*Be able to understand the natural environment and its relationships with human activities.*		
2	*Be able to apply the fundamental knowledge of science and engineering to assess environmental and health risk.*		
3	*Be able to understand environmental laws and regulations to develop guidelines and procedures for health and safety issues.*		
4	*Be able to solve scientific problem-solving related to air, water, noise & land pollution*		
Pre-Requisite:			
1	*Basic knowledge of Environmental science*		

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Unit	Content	Hrs/Unit	Marks/Unit
1	Basic ideas of environment, basic concepts, man, society & environment, their interrelationship (1L) Mathematics of population growth and associated problems, Importance of population study in environmental engineering, definition of resource, types of resource, renewable, non-renewable, potentially renewable, effect of excessive use vis-à-vis population growth, Sustainable Development. (2L) Materials balance: Steady state conservation system, steady state system with non-conservative pollutants, step function. (1L) Environmental degradation: Natural environmental Hazards like Flood, earthquake, Landslide-causes, effects and control/management; Anthropogenic	6	
	degradation like Acid rain-cause, effects and control. Nature and scope of Environmental Science and Engineering. (2L)		
2	Elements of ecology: System, open system, closed system, definition of ecology, species, population, community, definition of ecosystem- components types and function. (1L) Structure and function of the following ecosystem: Forest ecosystem, Grassland ecosystem, Desert ecosystem, Aquatic ecosystems, Mangrove ecosystem (special reference to Sundar ban); Food chain [definition and one example of each food chain], Food web.(2L) Biogeochemical Cycle- definition, significance, flow chart of different cycles with only elementary reaction [Oxygen, carbon, Nitrogen, Phosphate, Sulphur]. (1L) Biodiversity- types, importance, Endemic species, Biodiversity Hot-spot, Threats to biodiversity, Conservation of biodiversity.(2L)	**6**	

3	Atmospheric Composition: Troposphere, Stratosphere, Mesosphere, Thermosphere, Tropopause and Mesopause. (1L) Energy balance: Conductive and Convective heat transfer, radiation heat transfer, simple global temperature model [Earth as a black body, earth as albedo], Problems.(1L) Green house effects: Definition, impact of greenhouse gases on the global climate and consequently on sea water level, agriculture and marine food. Global warming and its consequence, Control of Global warming. Earth's heat budget.(1L) Lapse rate: Ambient lapse rate Adiabatic lapse rate, atmospheric stability, temperature inversion (radiation inversion).(2L) Atmospheric dispersion: Maximum mixing depth, ventilation coefficient, effective stack height, smokestack plumes and Gaussian plume model.(2L) Definition of pollutants and contaminants, Primary and secondary pollutants: emission standard, criteria	**11**	
	pollutant. Sources and effect of different air pollutants- Suspended particulate matter, oxides of carbon, oxides of nitrogen, oxides of sulphur, particulate, PAN. (2L) Smog, Photochemical smog and London smog. Depletion Ozone layer: CFC, destruction of ozone layer by CFC, impact of other green-house gases, effect of ozone modification. (1L) Standards and control measures: Industrial, commercial and residential air quality standard, control measure (ESP. cyclone separator, bag house, catalytic converter, scrubber (ventury), Statement with brief reference). (1L)		

4.	Hydrosphere, Hydrological cycle and Natural water. Pollutants of water, their origin and effects: Oxygen demanding wastes, pathogens, nutrients, Salts, thermal application, heavy metals, pesticides, volatile organic compounds. (2L) River/Lake/ground water pollution: River: DO, 5-day BOD test, Seeded BOD test, BOD reaction rate constants, Effect of oxygen demanding wastes on river [deoxygenation, reaeration], COD, Oil, Greases, pH. (2L) Lake: Eutrophication [Definition, source and effect]. (1L) Ground water: Aquifers, hydraulic gradient, ground water flow (Definition only)(1L) Standard and control: Waste water standard [BOD, COD, Oil, Grease], Water Treatment system [coagulation and flocculation, sedimentation and filtration, disinfection, hardness and alkalinity, softening] Waste water treatment system, primary and secondary treatments [Trickling filters, rotating biological contractor, Activated sludge, sludge treatment, oxidation ponds] tertiary treatment definition. (2L) Water pollution due to the toxic elements and their biochemical effects: Lead, Mercury, Cadmium, and Arsenic (1L)	**9**	
5	Lithosphere; Internal structure of earth, rock and soil (1L)	**3**	
	Solid Waste: Municipal, industrial, commercial, agricultural, domestic, pathological and hazardous solid wastes; Recovery and disposal method- Open dumping, Land filling, incineration, composting, recycling. Solid waste management and control (hazardous and biomedical waste).(2L)		

6	Definition of noise, effect of noise pollution, noise classification [Transport noise, occupational noise, neighbourhood noise] (1L) Definition of noise frequency, noise pressure, noise intensity, noise threshold limit value, equivalent noise level, L10 (18hr Index) ,n Ld.Noise pollution control. (1L)	**3**	
7	Environmental impactassessment, Environmental Audit, Environmental laws and protection act of India, Different international environmental treaty/ agreement/ protocol. (2L)	**2**	

Text books/ reference books:

1. M.P. Poonia & S.C. Sharma, Environmental Studies, Khanna Publishing House (AICTE Recommended Textbook – 2018)
2. Masters, G. M., "Introduction to Environmental Engineering and Science", *Prentice-Hall of India Pvt. Ltd.,1991.*
3. De, A. K., "Environmental Chemistry", New Age

International ***Course Outcomes:***

On completion of the course students will be able to

MC-401.1 To understand the natural environment and its relationships with human activities.

MC-401.2 To apply the fundamental knowledge of science and engineering to assess environmental and health risk.

MC-401.3 To develop guidelines and procedures for health and safety issues obeying the environmental laws and regulations.

MC-401.4 Acquire skills for scientific problem-solving related to air, water, noise& land pollution.

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Course: Object Oriented Programming with C++ Lab
Code:
Contact: 4P
Credit: 2

Name of the Course:	Object Oriented Programming with C++ Lab		
Course Code:		**Semester:**	V
Duration:	6 months	**Maximum Marks:**	100
Teaching Scheme:			
Theory:	0 hrs./week	Continuous Internal Assessment	
Tutorial:	NIL	**External Assessment:**	60
Practical:	4 hrs./week	**Distribution of marks:**	40
Credit Points:	2		

Unit 1.1:

Basic UNIX/Linux Operating System Commands & Editor [2P]
Introduction of UNIX/Linux Operating System which includes preliminary commands, start-up & shutdown methodology, file handling as well as introduction to editors like Vi editor.

Unit 1.2:

C & C++ Compiler [2P]
Introduction to GNU C & C++ compiler, as well as introduction to GNU & GDB script.

Unit 2.1:

C++ Language Basics, Control Statements [2P]
Introduction to C++, basic loop control, executing programs, writing functions, selection statements.

Unit 2.2:

Function & Recursion [2P]
Review of functions and parameters, command line arguments, recursion.

Unit 2.3:

IO, Pointer & String [4P]
I/O streams, arrays and string manipulation, pointers, structures & unions.

Unit 3.1:

Basics of OOPs and Constructors & Destructors [4P]
Object-Oriented Programming in C++, fundamentals of classes, constructors-destructors. Dealing with member functions.

Unit 3.2:

Polymorphism & Inheritance [6P]
Function & operator overloading and polymorphism (both static & dynamic), Dealing with inheritance, derived class handling, abstract class, virtual class, overriding.

Unit 4.1:

Template & Dynamic Memory Allocation [6P]
Template class, name-space & exception handling, Dynamic memory allocation, implementation of Linked Lists, using C++.

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Computer Organization & Architecture Lab
Subject Code: PCC-CSBS493
Credit:1.5

Objectives of the course:
This course will expose students to the following
1.To be able to understand the behavior logic gates, adder, decoder and multiplexer.
2. How Computer Systems work & the basic principles.

Detailed contents: All laboratory assignments are based on Hardware Description Language (VHDL or Verilog) Simulation

Module	**Content**
1	HDL introduction.
2	Basic digital logic based programming with HDL.
3	8-bit Addition, Multiplication, Division.
4	Design a BCD adder.
5	Design an 8×1 multiplexer.
6	Design a Decoder.
7	8-bit Register design.
8	Memory unit design and perform memory operations.
9	8-bit simple ALU design.
10	8-bit simple CPU design.
11	Interfacing of CPU and Memory.

Course Outcome:
1. Understand basic logic gates and efficiently verify the behavior of the circuit.
2. Acquire in-depth knowledge to implement different arithmetic operations.
3. Understand the working principle of the memory system.
4. Understand the working principle of CPU, and interfacing of CPU and Memory.

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Course Code	Course Title	L	T	P	J	C
ESC 481	**Design Thinking**	**0**	**10**	**3**	**0**	**2.5**
Pre-requisite	**NIL**					

Course Objectives:

1. Recognize the importance of design thinking and its various phases
2. Apply design thinking phases to create successful prototypes
3. Understand that both agile and design thinking process complement each other

Expected Course Outcome:

After the successful completion of the course the student should be able to

1. Understand the importance of design thinking and its different phases
2. Empathize with user situations and be able to define clear problem statements
3. Use the different ideation methods and come with different feasible and viable ideas for solving theproblem statements.
4. Create prototypes for clear understanding of the problem statement.
5. Test the created prototypes and be able to iterate if the design does not meet the customer requirement
6. Complement agile process with design thinking for efficient delivery process.

	Lab Experiments	
1	Immersion Activity	3 hours
2	Problem Definition	3 hours
3	Different Points of View	3 hours
4	Brainstorming session	3 hours
5	Drawing Mind Maps	3 hours
6	Ideation Games	3 hours
7	Creating Prototype	3 hours
8	Planning and working on video storyboard	3 hours
9	Completing the prototype as per schedule	3 hours
10	Testing the prototype	3 hours
	Total Lecture hours:	**30 hours**

Text Book(s)

1. Tim Brown, Change by Design: How Design Thinking Transforms Organizations and Inspires, 1st Edition, HarperCollins, 2009.
2. Eli Woolery, Design Thinking Handbook, Invision, 2019.

Reference Books

1. Nir Eyal , Hooked: How to build habit-forming, 2014
2. Rod Judkins, The Art of Creative Thinking, Sceptre; 1st edition, 2015.

5th Semester

Sl No	Types of Course	Code	Course Title	Title Hours per week			Credit
				L	**T**	**P**	
1	Professional Core Courses	PCC-CSBS 501	Computer Networks	3	1	0	4
2	Engineering Science Course	ESC(CSBS) 501	Software Engineering	3	1	0	3
3	Professional Core Courses	PCC-CSBS 502	Database Management Systems	3	1	0	3
4	Humanities &Social Sciences including Management courses	HSMC (CSBS)-501	Business Strategy	3	0	0	3
5	Engineering Science Course	ESC(CSBS) 502	Machine Learning	2	0	0	2
6	Professional Elective – I	PEC-CSBS- 501 A/B/C	Computer Graphics / Artificial Intelligence /E-Commerce and ERP	3	1	0	3
7	Mandatory Courses	MC- CSBS 501	Constitution of India/ Essence of Indian Knowledge Tradition	1	0	0	0
				16	3	0	18
Practical							
	Professional Core Courses	PCC-CSBS 591	Computer Networks	0	0	3	2
	Professional Core Courses	PCC-CSBS 592	Database Management Systems	0	0	3	2
	Engineering Science Course	ESC 591	Software Engineering	0	0	3	2
				0	0	2	6

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Computer Networks
Code: PCC-CSBS 501
Contact: 4 hrs

Name of the Course:	**Computer Networks**		
Course Code: PCC-CSBS501	Semester: V		
Duration:6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam:70 Marks	
Credit Points:	4		
Objective:			
1	To develop an understanding of modern network architectures from a design and performance perspective.		
2	To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs).		
3	To provide an opportunity to do network programming		
4	To provide a WLAN measurement ideas.		

Unit	Content	Hrs/Unit	Marks/Unit
1	Data communication Components: Representation of data and its flow Networks,Various Connection Topology, Protocols and Standards, OSI model, Transmission Media, LAN: Wired LAN, Wireless LANs, Connecting LAN and Virtual LAN, Techniques for Bandwidth utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum.	7	
2	Data Link Layer and Medium Access Sub Layer: Error Detection and Error Correction - Fundamentals, Block coding, Hamming Distance, CRC; Flow Control and Error control protocols - Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window, Piggybacking,	7	

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	Random Access, Multiple access protocols -Pure ALOHA, Slotted ALOHA,CSMA/CD,CDMA/CA		
3	Network Layer: Switching, Logical addressing – IPV4, IPV6; Address mapping – ARP, RARP, BOOTP and DHCP–Delivery, Forwarding and Unicast Routing protocols.	12	
4.	Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), SCTP Congestion Control; Quality of Service, QoS improving techniques: Leaky Bucket and Token Bucket algorithm.	7	
5	Application Layer: Domain Name Space (DNS), DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, Firewalls, Basic concepts of Cryptography.	7	

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Software Engineering

ESC(CSBS) 501

Contact : Contact: 4 hrs

Name of the Course:	**Software Engineering**
Course Code: **ESC(CSBS) 501**	Semester: V
Duration:6 months	Maximum Marks:100

Teaching Scheme	**Examination Scheme**
Theory:3 hrs./week	Mid Semester exam: 15
Tutorial: 1 hr./week	Assignment and Quiz: 10 marks
	Attendance: 5 marks
Practical: hrs./week	End Semester Exam:70 Marks

Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction:** Programming in the small vs. programming in the large; software project failures and importance of software quality and timely availability; of software engineering towards successful execution of large software projects; emergence of software engineering as a discipline, Software Engineering Historical Development from Jackson Structured Programming to Agile Development.	4	
2	**Software Project Management:** Basic concepts of life cycle models – different models and milestones; software project planning –identification of activities and resources; concepts of feasibility study; techniques for estimation of schedule and effort; software cost estimation models and concepts of software engineering economics; techniques of software project control and reporting; introduction to measurement of software size; introduction to the concepts of risk and its mitigation; configuration management.	6	
3	**Software Quality Management and Reliability:** Software quality; Garvin's quality dimensions, McCall's quality factor, ISO 9126 quality factor; Software Quality Dilemma; Introduction to Capability Maturity Models (CMM and CMMI); Introduction to software reliability, reliability models and estimation.	4	

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4	**Software Requirements Analysis, Design and Construction:** Introduction to Software Requirements Specifications (SRS) and requirement elicitation techniques; techniques for requirement modelling – decision tables, event tables, state transition tables, Petri nets; requirements documentation through use cases; introduction to UML, introduction to software metrics and metrics-based control methods; measures of code and design quality.	6	
5	**Object Oriented Analysis, Design and Construction:** Concepts -- the principles of abstraction, modularity, specification, encapsulation and information hiding; concepts of abstract data type; Class Responsibility Collaborator (CRC) model; quality of design; design measurements; concepts of design patterns; Refactoring; object-oriented construction principles; object oriented metrics.	6	
6	**Software Testing:** Introduction to faults and failures; basic testing concepts; concepts of verification and validation; black box and white box tests; white box test coverage – code coverage, condition coverage, branch coverage; basic concepts of black-box tests – equivalence classes, boundary value tests, usage of state tables; testing use cases; transaction based testing; testing for non-functional requirements – volume, performance and efficiency; concepts of inspection; Unit Testing, Integration Testing, System Testing and Acceptance Testing.	6	
7	**Agile Software Engineering:** Concepts of Agile Methods, Extreme Programming; Agile Process Model - Scrum, Feature; Scenarios and Stories	4	

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Database Management Systems
Code: PCC-CSBS 502

Contact: 4L

Name of the Course:	**Database Management Systems**		
Course Code: PCC-CSBS 502	Semester: V		
Duration:6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam:70 Marks	
Credit Points:	3		
Objective:			
1	To understand the different issues involved in the design and implementation of a database system.		
2	To study the physical and logical database designs, database modeling, relational, hierarchical, and network models		
3	To understand and use data manipulation language to query, update, and manage a database		
4	To develop an understanding of essential DBMS concepts such as: database security, integrity, concurrency, distributed database, and intelligent database, Client/Server (Database Server), Data Warehousing.		
5	To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.		
6	To understand the different issues involved in the design and implementation of a database system.		

Unit	Content	Hrs/Unit	Marks/Unit
1	**Database system architecture:** Data Abstraction, Data Independence,Data Definition Language(DDL),Data Manipulation Language(DML). **Data models:** Entity-relationship model, network model, relational and object oriented data models, integrity constraints, data manipulation operations.	9	

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2	**Relational query languages:** Relational algebra, Tuple and domain relational calculus, SQL3, DDL and DML constructs, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQLserver. **Relational database design:** Domain and data dependency, Armstrong's axioms, Normal forms, Dependency preservation, Losslessdesign. **Query processing and optimization:** Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization algorithms.	13	
3	**Storage strategies**: Indices, B-trees, hashing.	3	
4.	**Transaction processing:** Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp based schedulers, Multi-version and optimistic Concurrency Control schemes, Database recovery.	5	
5	**Database Security:** Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection.	3	
6	**Advanced topics:** Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.	3	

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Business Strategy
Code: HSMC-501
Contact: 3L

Name of the Course:	Business Strategy	
Course Code: HSMC-501	Semester: V	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: hrs./week		End Semester Exam:70 Marks
Credit Points:	3	

Objective:

1	To introduce the concepts of strategic management and understand its nature in competitive and institutional landscape.
2	To develop a holistic approach to see business issues comprehensively and using other core andfunctional subject knowledge for decision-making.
3	To identify and interpret the critical challenges and opportunities before an organization.
4	To introduce the concepts of strategic management and understand its nature in competitive and institutional landscape.

Course Out Come:

1. Learn the fundamental concepts of strategic management to analyze business situations and apply these
   concepts to solve business problems.
2. Understand the fundamental principles of and interrelationships among business functions such as:R&D, production, marketing, finance, HR and information technology
3. Understand the inter-relationships of business to individuals, other organizations, government andsociety.
4. Describe the tools of strategic analysis thoroughly, how they are used, and where they fit in the

managerial process to frame and implement strategies.

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Unit	Content	Hrs/Unit	Marks/Unit
1	Introduction: Importance of Strategic Management, Vision and Objectives, Schools of thought in Strategic Management, Strategy Content, Process, and Practice, Fit Concept and Configuration Perspective in StrategicManagement	6	
2	Internal Environment of Firm: Core Competence as the Root of Competitive Advantage, Sources of Sustained Competitive Advantage, Business Processes and Capabilities-based Approach to Strategy	6	
3	External Environment of Firm: Five Forces of Industry Attractiveness that Shape Strategy, The concept of Strategic Groups, and Industry Life Cycle	6	
4	Generic Strategies :Generic Strategies, Generic Strategies and the Value Chain	4	
5	**Corporate Strategy, and Growth Strategies** :The Motive for Diversification, Related and Unrelated Diversification, Business Portfolio Analysis	6	
6.	**Contesting with competitors in overseas markets** :Expansion, Integration and Diversification, Strategic Alliances, Joint Ventures, and Mergers & Acquisitions	6	
7.	**Strategy Implementation: Structure and Systems** :The 7S Framework, Strategic Control and Corporate Governance	4	

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Machine Learning
Code: ESC(CSBS) 502
Contact: 2L

Name of the Course:	**Machine Learning**		
Course Code: **ESC(CSBS) 502**	Semester: V		
Duration:6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:2 hrs./week		Mid Semester exam: 15	
Tutorial: NIL		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam:70 Marks	
Credit Points:	2		
Objective:			
1	Ability to comprehend the concept of supervised and unsupervised learning techniques		
2	Differentiate regression, classification and clustering techniques and to implement their algorithms.		
3	To analyze the performance of various machine learning techniques and to select appropriate featuresfor training machine learning algorithms.		
Course Out Come:			
1. Understand the concepts of various machine learning strategies. 2. Handle computational data and learn ANN learning models. 3. Solve real world applications by selecting suitable learning model. 4. Boost the performance of the model by combining results from different approaches. 5. Recognize and classify sequencing patterns using HMM. 6. Infer the association and relationship between the data objects. Construct machine learning model for unseen data and can solve real world application.			

Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction to Machine Learning:** Introduction to Machine Learning (ML); Feature engineering; Learning Paradigm, Generalization of hypothesis, VC Dimension, PAC learning, Applications of ML.	3	

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2	**Data Handling and ANN:** Feature selection Mechanisms, Imbalanced data, Outlier detection- Artificial neural networks including backpropagation- Applications	3	
3	**ML Models and Evaluation:** Regression: Multi-variable regression; Model evaluation; Least squares regression; Regularization; LASSO; Applications of regression, Classification – KNN, Naïve Bayes, SVM, Decision Tree; Training and testing classifier models; Cross-validation; Model evaluation (precision, recall, F1-mesure, accuracy, area under curve); Statistical decision theory including discriminant functions and decision surfaces	4	
4	**Model Assessment and Inference:** Model assessment and Selection – Ensemble Learning – Boosting, Bagging, Model Inference and Averaging, Bayesian Theory, EM Algorithm	4	
5	**Hidden Markov Models:** Hidden Markov Models (HMM) with forward-backward and Vierbi algorithms; Sequence classification using HMM; Conditional random fields; Applications of sequence classification such as part-of-speech tagging	3	
6.	**Association Rules:** Mining Association Rules in Large Databases. Mining Frequent Patterns-- basic concepts - Efficient and scalable frequent item set mining - methods, Apriori algorithm, FP-Growth algorithm	3	
7.	**Clustering:** K Means, Hierarchical Clustering – Single, complete, Average linkage; Ward's algorithm; Minimum spanning tree clustering; BIRCH clustering	4	

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Computer Graphics
Code: PEC-CSBS- 501 A
Contact: 4L

Name of the Course:	**Computer Graphics**	
Course Code: PEC-CSBS- 501 A	Semester: V	
Duration: 6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks
		Attendance : 5 marks
Practical: NIL		End Semester Exam :70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/ Unit
1	**Introduction to computer graphics & graphics systems [6L**]: Overview of computer graphics, representing pictures, preparing, presenting & interacting with pictures for presentations; Visualization & image processing; RGB color model, direct coding, lookup table; storage tube graphics display, Raster scan display, 3D viewing devices, Plotters, printers, digitizers, Light pens etc.; Active & Passive graphics devices; Computer graphics software. **Scan conversion [8L]:** Points & lines, Line drawing algorithms; DDA algorithm, Bresenham's line algorithm, Circle generation algorithm; Ellipse generating algorithm; scan line polygon, fill algorithm, boundary fill algorithm, flood fill algorithm.	14	
2	**2D transformation & viewing [15L]:** Basic transformations: translation, rotation, scaling; Matrix representations & homogeneous coordinates, transformations between coordinate systems; reflection shear; Transformation of points, lines, parallel lines, intersecting lines. Viewing pipeline, Window to view port co-ordinate transformation, clipping operations, point clipping, line clipping, clipping circles, polygons & ellipse. Cohenand Sutherland line clipping, Sutherland-Hodgeman Polygon clipping, Cyrus-beck clippingmethod 3D transformation & viewing [5L]: 3D transformations: translation, rotation, scaling & other transformations. Rotation about an arbitrary axis in space, reflection through an arbitrary plane; general parallel projection transformation; clipping, view port clipping, 3D viewing.	20	

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3	**Curves [3L]:** Curve representation, surfaces, designs, Bezier curves, B-spline curves, end conditions for periodic B-spline curves, rational B-spline curves. **Hidden surfaces [3L]:** Depth comparison, Z-buffer algorithm, Back face detection, BSP tree method, the Painter's algorithm, scan-line algorithm; Hidden line elimination, wire frame methods , fractal - geometry. **Color & shading models [2L]:** Light & color model; interpolative shading model; Texture. **Introduction to Ray-tracing: [3L]** Human vision and color, Lighting, Reflection and transmission models.	6	

Artificial Intelligence

Code: PEC-CSBS- 501 B

Contact: 4L

Name of the Course:	**Artificial Intelligence**	
Course Code**:** PEC-CSBS- 501 B	Semester: V	
Duration: 6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks
		Attendance : 5 marks
Practical: NIL		End Semester Exam :70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/ Unit
1	**Introduction [2]** Overview of Artificial intelligence- Problems of AI, AI technique, Tic - Tac - Toe problem. **Intelligent Agents [2]** Agents & environment, nature of environment, structure of agents, goal based agents, utility based agents, learning agents. **Problem Solving [2]** Problems, Problem Space & search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.	6	

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2.	**Search techniques [5]** Solving problems by searching :problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. **Heuristic search strategies [5]** Greedy best-first search, A* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search, **genetic algorithms; constraint satisfaction problems, local search for constraint satisfaction problems.** **Adversarial search [3]** **Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.**	13	
3	**Knowledge & reasoning [3]** **Knowledge representation issues, representation & mapping, approaches to knowledge representation, issues in knowledge representation.**	3	
4	**Using predicate logic [2]** **Representing simple fact in logic, representing instant & ISA relationship, computable functions & predicates, resolution, natural deduction.** **Probabilistic reasoning [4]** **Representing knowledge in an uncertain domain, the semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics.**	6	
5	**Natural Language processing [2]** **Introduction, Syntactic processing, semantic analysis, discourse & pragmatic processing.** **Learning [2]** **Forms of learning, inductive learning, learning decision trees, explanation based learning, learning using relevance information, neural net learning & genetic learning.** **Expert Systems [2]** **Representing and using domain knowledge, expert system shells, knowledge acquisition.**	6	

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E-Commerce & ERP

Code: PEC-CSBS- 501 C

Contacts: 4L

Name of the Course:	E-Commerce & ERP	
Course Code**:** PEC-CSBS- 501 C	Semester: V	
Duration: 6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks
		Attendance : 5 marks
Practical: NIL		End Semester Exam :70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/ Unit
1	Overview, Definitions, Advantages & Disadvantages of E – Commerce, Threats of E – Commerce, Managerial Prospective, Rules & Regulations For Controlling E – Commerce, CyberLaws. [3 L]	3	
2.	Technologies : Relationship Between E – Commerce & Networking, Different Types of Networking Commerce, Internet, Intranet & Extranet, EDI Systems Wireless Application Protocol : Definition, Hand Held Devices, Mobility & Commerce, Mobile Computing, WirelessWeb, Web Security, Infrastructure Requirement For E – Commerce . [5 L]	5	
3	Business Models of e – commerce : Model Based On Transaction Type, Model Based OnTransaction Party - B2B, B2C, C2B, C2C, E – Governance. [2 L]	2	
4	E – strategy : Overview, Strategic Methods for developing E – commerce. [2 L]	2	
5	Four C's : (Convergence, Collaborative Computing, Content Management & Call Center). Convergence : Technological Advances in Convergence – Types, Convergence and its implications, Convergence & Electronic Commerce. Collaborative Computing : Collaborative product development, contract as per	6	

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	CAD, Simultaneous Collaboration, Security. Content Management : Definition of content, Authoring Tools & Content Management, Content – partnership, repositories, convergence, providers, Web Traffic & Traffic Management ; Content Marketing. Call Center : Definition, Need, Tasks Handled, Mode of Operation, Equipment , Strength & Weaknesses of Call Center, Customer Premises Equipment (CPE). [6 L]		
	Supply Chain Management : E – logistics, Supply Chain Portal, Supply Chain Planning Tools(SCP Tools), Supply Chain Execution (SCE), SCE - Framework, Internet's effect on Supply Chain Power. [3 L]	3	
	E – Payment Mechanism : Payment through card system, E – Cheque, E – Cash, E – PaymentThreats & Protections. [1 L] E – Marketing :. Home –shopping, E-Marketing, Tele-marketing [1 L]	2	
	Electronic Data Interchange (EDI) : Meaning, Benefits, Concepts, Application, EDI Model,Protocols (UN EDI FACT / GTDI, ANSI X – 12), Data Encryption (DES / RSA). [2 L]	2	
	Risk of E – Commerce : Overview, Security for E – Commerce, Security Standards, Firewall, Cryptography, Key Management, Password Systems, Digital certificates, Digital signatures. [4 L]	4	
	Enterprise Resource Planning (ERP) : Features, capabilities and Overview of Commercial Software, re-engineering work processes for IT applications, Business Process Redesign, Knowledge engineering and data warehouse . Business Modules: Finance, Manufacturing (Production), Human Resources, Plant Maintenance, Materials Management, QualityManagement, Sales&Distribution ERPPackage, ERP Market: ERP Market Place, SAP AG, PeopleSoft, BAAN, JD Edwards, Oracle Corporation ERP-Present and Future: EnterpriseApplication Integration (EAI), ERP and E-Commerce, ERP and Internet, Future Directions in ERP [10]	10	

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Constitution of India
Code: MC-CSBS 501
Contact

Name of the Course:	**Constitution of India**	
Course Code: MC-CSBS 501	Semester: V	
Duration: 6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance : 5 marks
Practical: NIL		
Credit Points:	0	

Unit	Content	Hrs/Unit	Marks/ Unit
1.	**Introduction:** Constitution' meaning of the term,, Indian Constitution: Sources and constitutional history, Features: Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy	3	
2.	**Union Government and its Administration :** Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha	6	
3.	**State Government and its Administration Governor:** Role and Position, CM and Council of ministers, State Secretariat: Organisation, Structure and Functions	6	
4.	**Local Administration District's Administration head:** Role and Importance, Municipalities: Introduction, Mayor and roleof Elected Representative, CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: Zila Pachayat, Elected officials and their roles, CEO Zila Pachayat: Position and role, Block level: Organizational Hierarchy **(Different 4.departments), Village level:** Role of Elected and Appointed officials, Importance of grass root democracy	8	
5.	**Election Commission Election Commission:** Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women	2	

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PRACTICAL
SYLLABUS

Computer Networks Lab
Course Code: PCC- CSBS 591
Contact: 3P

Name of the Course:	Computer Networks Lab
Course Code: PCC-CSBS 591	Semester: V
Duration:6 months	Maximum Marks:100
Teaching Scheme:	
Theory: hrs./week	Continuous Internal Assessment
Tutorial: NIL	External Assesement:60
Practical: 3 hrs./week	Distribution of marks:40
Credit Points:	2

Laboratory Experiments:

1) NIC Installation & Configuration (Windows/Linux)
2) Understanding IP address, subnet etc

Familiarization with

- Networking cables (CAT5, UTP)
- Connectors (RJ45, T-connector)
- Hubs, Switches

3) TCP/UDP Socket Programming
   - Simple, TCP based, UDP based
   - Multicast & Broadcast Sockets
   - Implementation of a Prototype Multithreaded Server

Implementation of

4) Data Link Layer Flow Control Mechanism (Stop & Wait, Sliding Window) Data Link Layer Error Detection Mechanism (Cyclic Redundancy Check) Data Link Layer Error Control Mechanism (Selective Repeat, Go Back N)
   Server Setup/Configuration : FTP, TelNet,
5) NFS, DNS, Firewall

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Database Management System Lab

Code: PCC- CSBS 592

Name of the Course:	**Database Management System Lab**
Course Code: PCC-CSBS 592	Semester:V
Duration:6 months	Maximum Marks:100
Teaching Scheme:	
Theory: hrs./week	Continuous Internal Assessment
Tutorial: NIL	External Assesement:60
Practical: 3 hrs./week	Distribution of marks:40
Credit Points:	2

Laboratory Experiments:

Structured Query Language

1. **Creating Database**
   - Creating a Database
   - Creating a Table
   - Specifying Relational Data Types
   - Specifying Constraints
   - Creating Indexes

2. **Table and Record Handling**
   - INSERT statement
   - Using SELECT and INSERT together
   - DELETE, UPDATE, TRUNCATE statements
   - DROP, ALTER statements

3. **Retrieving Data from a Database**
   - The SELECT statement
   - Using the WHERE clause
   - Using Logical Operators in the WHERE clause
   - Using IN, BETWEEN, LIKE , ORDER BY, GROUP BY and HAVING

Clause

   - Using Aggregate Functions
   - Combining Tables Using JOINS
   - Subqueries

4. **Database Management**
   - Creating Views
   - Creating Column Aliases

- Creating Database Users
- Using GRANT and REVOKE

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Software Engineering
Lab Code: ESC591

Contact: 3P

Name of the Course:	**Software Engineering Lab**
Course Code: **ESC591**	Semester:V
Duration:6 months	Maximum Marks:100
Teaching Scheme:	
Theory: hrs./week	Continuous Internal Assessment
Tutorial: NIL	External Assesement:60
Practical: 3 hrs./week	Distribution of marks:40
Credit Points:	2
Course Outcomes:	
1	To understand the software engineering methodologies involved in the phases for project development.
2	To gain knowledge about open source tools used for implementing software engineering methods.
3	To exercise developing product-startups implementing software engineering methods.
4	Learn simple optimization techniques
Pre-Requisite:	
Laboratory Experiments:	

- Problem Analysis and Project Planning -Thorough study of the problem – Identify Project scope, Objectives and Infrastructure.
- Software Requirement Analysis – Describe the individual Phases/modules of the project and Identify deliverables. Identify functional and non-functionalrequirements.
- Data Modeling – Use work products – data dictionary.
- Software Designing - Develop use case diagrams and activity diagrams, build and testclass diagrams, sequence diagrams and add interface to class diagrams.
- Prototype model – Develop the prototype of theproduct.

The SRS and prototype model should be submitted for end semester examination.

Any experiment specially designed by the college
(Detailed instructions for Laboratory Manual to be followed for further guidance)

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Semester-VI

Design and Analysis of Algorithms
Code: PCC-CSBS601

Name of the Course:	**Design and Analysis of Algorithms**		
Course Code: PCC-CSBS601	Semester: VI		
Duration: 6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory: 3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam: 70 Marks	
Credit Points:	3		
Objective:			
1	The aim of this module is to learn how to develop efficient algorithms for simple computational tasks and reasoning about the correctness of them		
2	Through the complexity measures, different range of behaviors of algorithmsand the notion of tractable and intractable problems will be understood.		
Pre-Requisite:			
1	To know data-structure and basic programming ability		

Unit	Content	Hrs/Unit	Marks/Unit
1	Introduction: Characteristics of algorithm.Analysis of algorithm: Asymptotic analysis of complexity bounds – best, average and worst-case behavior; Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method andMasters' theorem	8	
2	Fundamental Algorithmic Strategies: Brute-Force, Greedy, Dynamic Programming, Branch and-Bound and Backtracking methodologies for the design of algorithms; Illustrations of these techniques for Problem-Solving, Bin Packing, Knap Sack TSP. Heuristics –characteristics and theirapplication domains.	8	
3	Graph and Tree Algorithms: Traversal algorithms:Depth First Search (DFS) and Breadth First Search(BFS); Shortest path algorithms, Transitiveclosure, Minimum Spanning Tree, Topological sorting, Network Flow Algorithm.	6	

	Tractable and Intractable Problems: Computability		
4.	of Algorithms, Computability classes – P,NP, NP-complete and NP-hard. Cook's theorem, Standard NP-complete problems and Reduction techniques.	10	
5	Advanced Topics: Approximation algorithms, Randomized algorithms, Class of problems beyond NP – P SPACE	4	

Text books/ reference books:

Introduction to Algorithms, 4TH Edition, Thomas H Cormen, Charles E Lieserson, RonaldL Rivest and Clifford Stein, MIT Press/McGraw-Hill.
Fundamentals of Algorithms – E. Horowitz et al.
Algorithm Design, 1ST Edition, Jon Kleinberg and ÉvaTardos, Pearson.
Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition,Michael T Goodrich and Roberto Tamassia, Wiley.
Algorithms -- A Creative Approach, 3RD Edition, UdiManber, Addison-Wesley, Reading,MA
Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House (AICTE Recommended Textbook – 2018)
Algorithms Design and Analysis, Udit Agarwal, Dhanpat Rai

Course Outcomes

On completion of the course students will be able to
PCC-CSBS601.1 For a given algorithms analyze worst-case running times of algorithmsbased on asymptotic analysis and justify the correctness of algorithms.
PCC-CSBS601.2 Describe the greedy paradigm and explain when an algorithmic designsituation calls for it. For a given problem develop the greedy algorithms.
PCC-CSBS601.3 Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Synthesize divide-and-conquer algorithms. Deriveand solve recurrence relation.
PCC-CSBS601.4 Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. For a given problems of dynamic-programming and
PCC-CSBS601.5 develop the dynamic programming algorithms, and analyze it to determineits computational complexity.
PCC-CSBS601,6 For a given model engineering problem model it using graph and write thecorresponding algorithm to solve the problems.
PCC-CSBS601.7 Explain the ways to analyze randomized algorithms (expected runningtime, probability of error).
PCC-CSBS601.8 Explain what an approximation algorithm is. Compute the approximationfactor of an approximation algorithm (PTAS and FPTAS).

Operating Systems
Code: PCC-CSBS602

Name of the Course:	**Operating Systems**		
Course Code: PCC-CSBS602	Semester: VI		
Duration: 6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr/week		Assignment and Quiz: 10 marks	
		Attendance : 5 marks	
Practical: hrs./week		End Semester Exam :70 Marks	
Credit Points:	3		
Objective:			
1	To learn the mechanisms of OS to handle processes and threads and their communication		
2	To learn the mechanisms involved in memory management in contemporary OS		
3	To gain knowledge on distributed operating system concepts that includes architecture,Mutual exclusion algorithms, deadlock detection algorithms and agreement protocols		
4	To know the components and management aspects of concurrency management		
Pre-Requisite:			

1	Computer Organization &Architecture

Unit	Content	Hrs/Unit	Marks/ Unit
1	**Introduction:** Concept of Operating Systems,Generations of Operating systems, Types of Operating Systems, OS Services, System Calls,Structure of an OS - Layered, Monolithic, Microkernel Operating Systems, Concept of Virtual Machine. Case study on UNIX and WINDOWS Operating System.	3	

2	**Processes:** Definition, Process Relationship, Different states of a Process, Process State transitions, Process Control Block (PCB), Context switching **Thread:** Definition, Various states, Benefits of threads, Types of threads, Concept of multithreads, **Process Scheduling**: Foundation and Schedulingobjectives, Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time; Scheduling algorithms: Pre-emptive and Non pre-emptive, FCFS, SJF, RR; Multiprocessor scheduling: Real Time scheduling: RM and EDF.	10	
3.	**Inter-process Communication:** Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Strict Alternation, Peterson's Solution, The Producer Consumer Problem, Semaphores, Event Counters, Monitors, Message Passing, Classical IPC Problems:Reader's & Writer Problem, Dinning Philosopher Problemetc.	5	
4.	**Deadlocks:** Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.	5	
5.	**Memory Management:** Basic concept, Logical andPhysical address map, Memory allocation: Contiguous Memory allocation– Fixed and variable partition– Internal and External fragmentation and Compaction; Paging: Principle of operation –Page allocation Hardware support for paging, Protection and sharing, Disadvantages of paging. **Virtual Memory**: Basics of Virtual Memory – Hardware and control structures – Locality ofreference, Page fault , Working Set , Dirty page/Dirtybit – Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU) and Least Recently	8	

	used(LRU).		

6.	**I/O Hardware:** I/O devices, Device controllers, Direct memory access Principles of I/O Software: Goals ofInterrupt handlers, Device drivers, Device independent I/O software, Secondary-Storage Structure: Disk structure, Disk scheduling algorithms **File Management**: Concept of File, Access methods, File types, File operation, Directory structure, File System structure, Allocation methods (contiguous, linked, indexed), Free-space management (bit vector,linked list, grouping), directory implementation (linear list, hash table), efficiency andperformance. **Disk Management:** Disk structure, Disk scheduling -FCFS, SSTF, SCAN, C-SCAN, Disk reliability, Diskformatting, Boot-block, Bad blocks	6	

Text book and Reference books:

Operating System Concepts Essentials, 9th Edition by AviSilberschatz,Peter Galvin, Greg Gagne, Wiley Asia Student Edition.
Operating Systems: Internals and Design Principles, 5th Edition, WilliamStallings, Prentice Hall of India.
Operating System Concepts, Ekta Walia, Khanna Publishing House(AICTE Recommended Textbook – 2018)
Operating System: A Design-oriented Approach, 1st Edition by CharlesCrowley, Irwin Publishing
Operating Systems: A Modern Perspective, 2nd Edition by Gary J. Nutt, Addison-Wesley
Design of the Unix Operating Systems, 8th Edition by Maurice Bach,Prentice-Hall of India
Understanding the Linux Kernel, 3rd Edition, Daniel P. Bovet, MarcoCesati, O'Reilly and Associates

Course Outcomes:
On completion of the course students will be able to
Create processes and threads.
Develop algorithms for process scheduling for a given specification of CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time.
For a given specification of memory organization develop the techniquesfor optimally allocating memory to processes by increasing memory utilization and for improving the access time. Design and implement filemanagement system.

For a given I/O devices and OS (specify) develop the I/O management functions in OS as part of a uniform device abstraction by performing operations for synchronization between CPU and I/O controllers.

Pattern Recognition
Code: ESC-CSBS601

Name of the Course:	**Pattern Recognition**		
Course Code: ESC-CSBS601	Semester: VI		
Duration:6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr/ week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: NIL		End Semester Exam:70 Marks	
Credit Points:	3		

Unit	Content	Hrs/Unit	Marks/Unit
1	Basics of pattern recognition	2	
2	**Bayesian decision theory 8L** Classifiers, Discriminant functions, Decisionsurfaces Normal density and discriminant functionsDiscrete features	8	
3	**Parameter estimation methods 6L** Maximum-Likelihood estimation Gaussian mixture models Expectation-maximization method Bayesian estimation	6	
4.	**Hidden Markov models for sequential patternclassification 8L** Discrete hidden Markov models Continuous density hidden Markovmodels	8	
5	**Dimension reduction methods 3L** Fisher discriminant analysis, Principal component analysis, Parzen-window method, K-Nearest Neighbour method	3	
6	Non-parametric estimation techniques for Density estimation	2	

7	**Linear discriminant function based classifier 5L** Perceptron Support vector machines	5	
8	**Non-metric methods for pattern classification 4L** Non-numeric data or nominal dataDecision trees	4	
9	**Unsupervised learning and clustering 2L** Criterion functions for clustering Algorithms for clustering: K-means, Hierarchical and other methods	2	

Text book and Reference books:

1. R. O. Duda, P. E. Hart and D. G. Stork: Pattern Classification, John Wiley, 2001.
2. S. Theodoridis and K. Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009.
3. C. M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.

Information Security
Code: PEC-CSBS601 A

Name of the Course:	Information Security		
Course Code: PEC-CSBS601A	Semester: VI		
Duration: 6 months	Maximum Marks: 100		
Teaching Scheme		**Examination Scheme**	
Theory: 3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr/Week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: NIL		End Semester Exam: 70 Marks	
Credit Points:	3		
Objective:			
1	To develop an understanding of modern network architectures from a design and performance perspective.		
2	To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs) and Wireless LANs (WLANs).		
3	To provide an opportunity to do network programming		
4	To provide a WLAN measurement ideas.		

Unit	Content	Hrs/Unit	Marks/Unit

1	Introduction: Introduction to Cyber Security, Importance and challenges in Cyber Security, Cyberspace, Cyber threats, Cyberwarfare, CIA Triad, Cyber Terrorism, Cyber Security of CriticalInfrastructure, Cybersecurity - Organizational Implications.	6	
2	Hackers and Cyber Crimes: Types of Hackers, Hackers and Crackers, Cyber-Attacks and Vulnerabilities, Malware threats, Sniffing, Gaining Access, Escalating Privileges, Executing Applications, Hiding Files, Covering Tracks, Worms, Trojans, Viruses, Backdoors.	7	
3	Ethical Hacking and Social Engineering: Ethical Hacking Concepts and Scopes, Threats and Attack Vectors, Information Assurance, Threat Modelling, Enterprise Information Security Architecture, Vulnerability Assessment and Penetration Testing, Types of Social Engineering, Insider Attack, Preventing Insider Threats, Social Engineering Targets and Defence Strategies.	8	
4.	Cyber Forensics and Auditing: Introduction to CyberForensics, Computer Equipment and associated storage media, Role of forensics Investigator, Forensics Investigation Process, Collecting Networkbased Evidence, Writing Computer Forensics Reports, Auditing, Plan an audit against a set of audit criteria, Information Security Management System Management. Introduction to ISO 27001:2013	10	
5	Cyber Ethics and Laws: Introduction to Cyber Laws,E-Commerce and E-Governance, Certifying Authority and Controller, Offences under IT Act, Computer Offences and its penalty under IT Act 2000, Intellectual Property Rights in Cyberspace. at Network Layer-IPSec.	5	

Text book and Reference books:

1. Cyber security , Nina Gobole & Sunit Belapune; Pub: Wiley India.
2. Information Security and Cyber Laws, Pankaj Agarwal
3. Donaldson, S., Siegel, S., Williams, C.K., Aslam, A., Enterprise Cybersecurity -How toBuild a Successful Cyberdefense Program

Against Advanced Threats, A-press
4. Nina Godbole, SumitBelapure, Cyber Security, Willey
5. Hacking the Hacker, Roger Grimes, Wiley
6. Cyber Law By Bare Act, Govt Of india, It Act 2000.
7. Information Security & Cyber Laws, Gupta & Gupta, Khanna Publishing House, (AICTERecommended Textbook- 2018)

Human Computer Interaction
Code: PEC-CSBS601B

Name of the Course:	**Human Computer Interaction**		
Course Code: PEC-CSBS601B	Semester: VI		
Duration: 6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory:3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr/Week		Assignment and Quiz: 10 marks	
		Attendance : 5 marks	
Practical: NIL		End Semester Exam :70 Marks	
Credit Points:	3		
Objective:			
1	Learn the foundations of Human Computer Interaction		
2	Be familiar with the design technologies for individuals and persons with disabilities		
3	Be aware of mobile Human Computer interaction		
4	Learn the guidelines for user interface.		
Pre-Requisite:			
1	Computer Organization &Architecture		

Unit	Content	Hrs/ Unit	Marks /Unit
1	Human: I/O channels – Memory – Reasoning and problem solving;The computer: Devices – Memory – processing and networks;	9	
	Interaction: Models – frameworks – Ergonomics – styles – elements –interactivity- Paradigms.		

2	Interactive Design basics – process – scenarios – navigation – screendesign – Iteration and prototyping. HCI in software process – software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules. Evaluation Techniques –Universal Design.	11	
3.	Cognitive models –Socio-Organizational issues and stake holder requirements –Communication and collaboration models- Hypertext,Multimedia and WWW.	8	
4.	Mobile Ecosystem: Platforms, Application frameworks- Types ofMobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.	8	
5.	Designing Web Interfaces – Drag & Drop, Direct Selection, ContextualTools, Overlays, Inlays and Virtual Pages, Process Flow. CaseStudies.	8	
6.	Recent Trends: Speech Recognition and Translation, Multimodal System	3	

Text book and Reference books:

1. Theodor Richardson, Charles N Thies, Secure Software Design, Jones & Bartlett
2. Kenneth R. van Wyk, Mark G. Graff, Dan S. Peters, Diana L. Burley, Enterprise SoftwareSecurity,
Addison Wesley.

Course Outcomes:

On completion of the course students will be able to

1. Differentiate between various software vulnerabilities.
2. Software process vulnerabilities for an organization.
3. Monitor resources consumption in a software.
4. Interrelate security and software development process.

Cloud Computing
Code: PEC-CSBS601C

Name of the Course:	**Cloud Computing**		
Course Code: PEC-CSBS601C	Semester: VI		
Duration: 6 months	Maximum Marks: 100		
Teaching Scheme		**Examination Scheme**	
Theory: 3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr/ Week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical:		End Semester Exam: 70 Marks	
Credit Points:	3		

Unit	Content	Hrs/Unit	Marks/Unit
1	**Definition of Cloud Computing and itsBasics (Lectures)**. Defining a Cloud, Cloud Types – NIST model, Cloud Cube	9	
	model, Deployment models (Public , Private, Hybrid and Community Clouds), Service Platform as a Service, Software as a Service with examples of services/ service providers, models – Infrastructure as a Service, Cloud Reference model, Characteristics of Cloud Computing – a shift in paradigm Benefits and advantages of Cloud Computing, A brief introduction on Composability, Infrastructure, Platforms, Virtual Appliances, Communication Protocols, Applications, Connecting to the Cloud by Clients, IaaS – Basic concept, Workload, partitioning ofvirtual private server instances, Pods, aggregations, silos PaaS – Basic concept, tools and development environment with examples SaaS - Basic concept and characteristics,Open SaaS and SOA, examples of SaaSplatform Identity as a Service (IDaaS) Compliance as a Service (CaaS)		

2	**Use of Platforms in Cloud Computing** Concepts of Abstraction and Virtualization Virtualization technologies : Types of virtualization (access, application, CPU,storage), Mobility patterns (P2V, V2V, V2P,P2P, D2C, C2C, C2D, D2D) Load Balancingand Virtualization: Basic Concepts, Networkresources for load balancing, Advanced loadbalancing (including Application DeliveryController and Application Delivery Network),Mention of The Google Cloud as an exampleof use of load balancing Hypervisors: Virtual machine technology and types, VMware vSphere Machine Imaging (including mentionof Open Virtualization Format – OVF) Porting of applications in the Cloud: The simple Cloud API and AppZero Virtual Application appliance, Concepts of Platform as a Service, Definition of services, Distinction between SaaS and PaaS (knowledge of Salesforce.com and Force.com), Application development Use of PaaS Application frameworks,	12	

	Discussion of Google Applications Portfolio – Indexed search, Dark Web, Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service., Discussion of Google Applications Portfolio – Indexed search, Dark Web, Aggregation and disintermediation, Productivity applications and service, Adwords, Google Analytics, Google Translate, a brief discussion on Google Toolkit (including introduction of Google APIs in brief), major features of Google App Engine service, Windows Azure platform: Microsoft's approach, architecture, and main elements, overview of Windows Azure AppFabric, Content Delivery Network, SQL Azure,		

	and Windows Live services,		
3	**<u>Cloud Infrastructure</u>**: Cloud Management: An overview of the features of network management systems and a brief introduction of related products from large cloud vendors, Monitoring of an entire cloud computing deployment stack – an overview with mention of some products, Lifecycle management ofcloud services (six stages of lifecycle). Concepts of Cloud Security: Cloud security concerns, Security boundary, Security service boundary Overview of security mapping Security of data: Brokered cloud storage access, Storage location and tenancy, encryption, and auditing and compliance Identity management (awareness of Identity protocol standards)	7	
4.	**<u>Concepts of Services and Applications :</u>** Service Oriented Architecture: Basic conceptsof message-based transactions, Protocol stack for an SOA architecture, Event-driven SOA, Enterprise Service Bus, Service catalogs, Applications in the Cloud: Concepts of cloudtransactions, functionality mapping,	8	
	Application attributes, Cloud service attributes, System abstraction and CloudBursting, Applications and Cloud APIs Cloud-based Storage: Cloud storage definition – Manned and Unmanned Webmail Services: Cloud mail services including Google Gmail, Mail2Web, WindowsLive Hotmail, Yahoo mail, concepts of Syndication services		

Text book and Reference books:

1. Cloud Computing Bible by Barrie Sosinsky, Wiley India Pvt. Ltd, 2013
2. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, McGraw Hill Education (India) Private Limited,2013
3. Cloud computing: A practical approach, Anthony T. Velte, Tata Mcgraw-Hill
4. Cloud Computing, Miller, Pearson
5. Building applications in cloud: Concept, Patterns and Projects, Moyer, Pearson
6. Cloud Computing – Second Edition by Dr. Kumar Saurabh, Wiley India

Data Mining & Analytics
Code: PEC-CSBS601D

Name of the Course:	**Data Mining & Analytics**	
Course Code: PEC-CSBS601D	Semester: VI	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory: 3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr/ Week		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical:		End Semester Exam: 70 Marks
Credit Points:	3	

PURPOSE	To acquire knowledge of Data mining techniques
At the end of the course, students will be able to	
1	Understand the concepts of Data Mining
2	Familiarize with association rule mining
3	Familiarize various classification algorithms
4	Understand the concepts of Cluster analysis
5	Implement the Data mining concepts with various domains

Session	**Description of Topic**	**Contact hours**
	UNIT I: Introduction	9
1	Introduction to Data Mining – Kinds of Data	2
2	Data mining Functionalities – Interesting Patterns	2
3	Task Primitives	1
4	Issues in Data Mining	1
5	Data Preprocessing	3

	UNIT II: Association Rules	8
6	Basic Concepts	1
7	Frequent Item Set Mining Methods	3
8	Association Rules	2
9	Correlation analysis	2
	UNIT III: Classification and Prediction	9
10	Issues Regarding Classification and Prediction	1
11	Decision Tree Induction Classification	2
12	Bayesian and Rule Based Classification	3
13	Support Vector Machine	2
14	Prediction	1
	UNIT IV: Cluster Analysis	9
15	What is Cluster Analysis	1
16	Types of Data in Cluster Analysis	2
17	Categorization of Clustering Methods	3
18	Hierarchical Methods	3
	UNIT V: PLASTIC ANALYSIS	10
19	Applications and Trends in Data Mining	3
20	Machine learning	3
21	Big data	2
22	Cloud computing	2
	Total contact hours	45

LEARNING RESOURCES

Sl. No.	**TEXT BOOKS**
1.	Jiawei Han and Micheline Kamber,"Data Mining – Concepts and Techniques", Second Edition, Morgan Kaufmann Publishers, 2006.
REFERENCE BOOKS/OTHER READING MATERIAL	
2.	M. H. Dunham, "Data Mining: Introductory and Advanced Topics", Pearson Education. 2001.
3.	D. Hand, H. Mannila and P. Smyth, "Principles of Data Mining", Prentice Hall. 2001.
4.	I H. Witten and E. Frank, "Data Mining: Practical Machine Learning Tools and Techniques", Morgan Kaufmann. 2000.

5.	Nathan Marz, James Warren, “Big Data-Principles and best practices of scalable real-time data systems”, DreamTech Press, 2015
6.	Arshdeep Bahga, Vijay Madisetti, "Cloud Computing: A Hands-On Approach", University Press, 2016

Compiler Design
Code: PEC-CSBS602A

Name of the Course:	**Compiler Design**	
Course Code: PEC-CSBS602A	Semester: VI	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr/Week		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam:70 Marks
Credit Points:	3	

Objective:	
1	To understand and list the different stages in the process of compilation.
2	Identify different methods of lexical analysis
3	Design top-down and bottom-up parsers
4	Identify synthesized and inherited attributes
5	Develop syntax directed translation schemes
6	Develop algorithms to generate code for a target machine

Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction to Compiling** [3L] Compilers, Analysis of the source program, The phases of the compiler, Cousins of the compiler.	3	
2	**Lexical Analysis** [6L] The role of the lexical analyzer, Tokens, Patterns, Lexemes, Input buffering, Specifications of a token, Recognition of a tokens, Finite automata, From a regular expressionto an NFA, From a regular expression to NFA, From a regular expression to DFA, Design of a lexical analyzer generator (Lex).	6	

3	**Syntax Analysis** [9L] The role of a parser, Context free grammars, Writing a grammar, Top down Parsing, Non-recursive Predictive parsing (LL), Bottom up parsing, Handles, Viable prefixes, Operator precedence parsing, LR parsers (SLR, LALR), Parser generators (YACC). Error Recoverystrategies for different parsing techniques.	9	
4	**Syntax directed translation** [5L] Syntax director definitions, Construction of syntaxtrees, Bottom-up evaluation of S attributed definitions, L attributed definitions, Bottom-up evaluation of inherited attributes.	5	
5	**Type checking** [4L] Type systems, Specification of a simple type checker, Equivalence of type expressions, Type conversions	4	
6	**Run time environments** [5L] Source language issues (Activation trees, Controlstack, scope of declaration, Binding of names), Storage organization (Subdivision of run-time memory, Activation records), Storage allocation strategies, Parameterpassing (call by value, call by reference, copy restore, call by name), Symbol tables, dynamic storage allocation techniques.	5	
7	**Intermediate code generation** [4L] Intermediate languages, Graphical representation,Three-address code, Implementation of three address statements (Quadruples, Triples, Indirect triples).	4	
8	**Code optimization** [5L] Introduction, Basic blocks & flow graphs, Transformation of basic blocks, Dag representationof basic blocks, The principle sources of optimization, Loops in flowgraph, Peephole optimization.	5	
9	**Code generations** [4L] Issues in the design of code generator, a simple code generator, Register allocation & assignment.	4	

Text book and Reference books:

1. Aho, Sethi, Ullman - "Compiler Principles, Techniques and Tools" - Pearson Education.
2. Holub - "Compiler Design in C" - PHI.

Course Outcomes:

On completion of the course students will be able to

1. Understand given grammar specification develop the lexical analyser
2. Design a given parser specification design top-downand bottom-up parsers
3. Develop syntax directed translation schemes
4. Develop algorithms to generate code for a target machine

Image Processing
Code: PEC-CSBS602 B

Name of the Course:	Image Processing	
Course Code: PEC-CSBS602 B	Semester: VI	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr/ Week		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam:70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction [3L]** Background, Digital Image Representation, Fundamental steps in Image Processing, Elements of Digital Image Processing - Image Acquisition, Storage, Processing, Communication, Display.	9	
2	**Digital Image Formation [4L]** A Simple Image Model, Geometric Model- Basic Transformation (Translation, Scaling, Rotation), Perspective Projection, Sampling & Quantization -Uniform & Non uniform.	4	
3	**Mathematical Preliminaries[9L]** Neighbour of pixels, Connectivity, Relations, Equivalence & Transitive Closure; Distance Measures, Arithmetic/Logic Operations, Fourier Transformation, Properties of The Two Dimensional Fourier Transform, Discrete Fourier Transform, Discrete Cosine & SineTransform.	9	

4.	**Image Enhancement [8L]** Spatial Domain Method, Frequency Domain Method, Contrast Enhancement -Linear & Nonlinear Stretching, Histogram Processing; Smoothing - Image Averaging, Mean Filter, Low-pass Filtering; Image Sharpening. High- pass Filtering, High-boost Filtering, Derivative Filtering, Homomorphic Filtering; Enhancement in the frequency domain - Low pass filtering, High pass filtering.	8	
5	**Image Restoration [7L]** Degradation Model, Discrete Formulation, Algebraic Approach to Restoration - Unconstrained & Constrained; Constrained Least Square Restoration, Restoration by Homomorphic Filtering, Geometric Transformation - Spatial Transformation, Gray Level Interpolation.	7	
6	**Image Segmentation [7L]** Point Detection, Line Detection, Edge detection, Combined detection, Edge Linking & Boundary Detection - Local Processing, Global Processing via The Hough Transform; Thresholding - Foundation, Simple Global Thresholding, Optimal Thresholding; Region Oriented Segmentation - Basic Formulation, Region Growing by Pixel Aggregation, Region Splitting & Merging.	7	

Text book and Reference books:

1. Hearn, Baker – "Computer Graphics (C version 2nd Ed.)" – Pearson education
2. Z. Xiang, R. Plastock – " Schaum's outlines Computer Graphics (2nd Ed.)" – TMH
3. D. F. Rogers, J. A. Adams – "Mathematical Elements for Computer Graphics (2nd Ed.)" –TMH

Robotics and Embedded Systems
Code: PEC-CSBS602 C

Name of the Course:	Robotics and Embedded Systems	
Course Code: PEC-CSBS602 C	Semester: VI	
Duration: 6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory: 3 hrs./week		Mid Semester exam: 15
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: hrs./week		End Semester Exam: 70 Marks
Credit Points:	3	

Objective:

1	In engineering courses students read subjects like; C/C++, Basic Electronics/Electrical, different types Sensors and Actuators, Microcontrollers and Microprocessors, different types of communication protocols and about many more. But they do not get scope to use that knowledge during their course. This course is especially designed to bridge that gap by providing an opportunity to the students, so that they can write embedded C/C++ programs to interface different types of input/output devices with the Microcontroller to do different projects. Now robotics is an emerging field of technology. In many sectors in our industry, robots are replacing humans very rapidly. That is why in this course students will also get some insight of robotics.

Course outcome

After completion of the training, students will able to:

- Understand the importance of embedded systems and robotics in our daily life.
- Identify different embedded devices.
- Co-related embedded systems with their university courses.
- Identify different components of embedded systems and robotics.
- Know about different features of a microcontroller.
- Write embedded C/C++ programs in different embedded systems programming platforms.
- Interfaced different input/output devices with a microcontroller.
- Design mechanical structure of a robot.
- Understand the robot configuration and sub-systems
- Interface different components of robot with microcontroller.
- Understand principle of robot programming.

	• Design different types of robots for different purposes. • Design wide varieties of embedded systems projects. • Do their Diploma/B Tech projects themselves.

Unit	Content	Hrs/Unit	Marks/Unit
1	AVR Microcontroller Introduction to AVR Microcontroller, Mega AVR Microcontroller series, Introduction to ATmeg16/32, Features, Architecture, Pin configurations, I/O ports, Port operation registers, Interrupts, ADC, Timers/counters, SPI, USART, Memory programming, etc.	10	
2	Embedded C/C++ Introduction to C/C++, Use of Loops, Array, Function, etc in C/C++, Introduction to Embedded C/C++ platform like; Atmel Studio and Proteus, Introduction to port operation registers programming, Programming to interface LED with ATmega16, etc.	10	
3	Robotics – Interfacing of Sensors, Motors, Display devices, etc : Introduction concept and mechanism of Robotics, Applications of Robotics, Introductions to Robotics components like; Motors, Sensors, Display devices, etc, Programming and interfacing of DC Motors, Stepper Motor, Servo Motors, Sensors (Analog & Digital), LCD, Communications modules like; Bluetooth, Xbee, etc.	8	
	Application: Digital notice board, Object counter, Digital temperature monitoring system, Range finder, Project using external interrupts, Stopwatch, Velocity control of DC Motor, Line follower Robot, Object avoider Robot, Intelligent home automation system, Solar seeker Robot, Robot communication using Bluetooth, RF Module, Xbee module, etc.	8	

SOFTWARE DESIGN USING UML
Code: PEC-CSBS602D

Name of the Course:	SOFTWARE DESIGN USING UML		
Course Code: PEC-CSBS602D	Semester: VI		
Duration: 6 months	Maximum Marks:100		
Teaching Scheme		**Examination Scheme**	
Theory: 3 hrs./week		Mid Semester exam: 15	
Tutorial: 1 hr./week		Assignment and Quiz: 10 marks	
		Attendance: 5 marks	
Practical: hrs./week		End Semester Exam: 70 Marks	
Credit Points:	3		

Objective:

1	To understand the fundamentals of object modeling • To understand and differentiate Unified Process from other approaches • To design with static UML diagrams. • To design with the UML dynamic and implementation diagrams • To improve the software design with design patterns. • To test the software against its requirements specification

Course outcome

	Upon Completion of the course, the students should be able to: • Express software design with UML diagrams • Design software applications using OO concepts. • Identify various scenarios based on software requirements • Transform UML based software design into pattern based design using design patterns • Understand the various testing methodologies for OO software

Unit	Content	Hrs/Unit	Marks/Unit
1	**UNIFIED PROCESS AND USE CASE DIAGRAMS** Introduction to OOAD with OO Basics - Unified Process – UML diagrams – Use Case –Case study – the Next Gen POS system, Inception -Use case Modelling – Relating Use cases – include, extend and generalization – When to use Use-cases.	**8**	
2	**STATIC UML DIAGRAMS** Class Diagram–– Elaboration – Domain Model – Finding conceptual classes and description classes – Associations – Attributes – Domain model refinement – Finding conceptual class Hierarchies – Aggregation and Composition - Relationship between sequence diagrams and use cases – When to use Class Diagrams. .	8	
3	**DYNAMIC AND ARCHITECTURAL MODELING UML DIAGRAMS** Dynamic Diagrams – UML interaction diagrams - System sequence diagram – Collaboration diagram – When to use Communication Diagrams - State machine diagram and Modelling – When to use State Diagrams - Activity diagram – When to use activity diagrams Implementation Diagrams - UML package diagram - When to use package diagrams - Component and Deployment Diagrams – When to use Component and Deployment diagrams.	8	
4	**DESIGN PATTERNS AND ELEMENTS DESIGN PATTERNS** GRASP-Designing objects with responsibilities – Applying GoF design patterns – Creational Patterns, Structural Patterns , Behavioral Patterns, Design Elements: Architectural design elements - Interface design elements - Component level diagram elements - Deployment level design elements, Mapping design to code.	8	

5.	**AGILE METHODOLOGY** Theories for Agile Management – Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agility in Design, Testing – Agile Documentations – Agile Drivers, Capabilities and Values	8	

Maulana Abul Kalam Azad University of Technology, West Bengal
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(Applicable from the Academic Session 2020-2021)

Big Data Analytics

Code: OEC-CSBS601A
Contacts: 3L

Name of the Course:	**Big Data Analytics**	
Course Code :OEC-CSBS601A	Semester:VI	
Duration:6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory: 3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz : 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam: 70 Marks
Credit Points:	3	

Total Number of Lectures: 48

COURSE OBJECTIVE	
□ Understand big data for business intelligence. Learn business case studies for big data analytics. Understand nosql big data management. Perform map-reduce analytics using Hadoop and related tools	
LECTURE WITH BREAKUP	**NO. OF LECTUR**
Unit 1: What is big data, why big data, convergence of key trends, unstructured data, industry examples of big data, web analytics, big data and marketing, fraud and big data, risk and big data, credit risk management, big data and algorithmic trading, big data and healthcare, big data in medicine, advertising and big data, big data technologies, introduction to Hadoop, open source technologies, cloud and big data, mobile business intelligence, Crowd sourcing analytics, inter and trans firewall analytics.	8
Unit 2: Introduction to NoSQL, aggregate data models, aggregates, key-value and document data models, relationships, graph databases, schemaless databases, materialized views, distribution models, sharding, master-slave replication, peer-peer replication, sharding and replication, consistency, relaxing consistency, version stamps, map-reduce, partitioning and combining, composing map-reduce calculations.	8
Unit 3: Data format, analyzing data with Hadoop, scaling out, Hadoop streaming, Hadoop pipes, design of Hadoop distributed file system (HDFS), HDFS concepts, Java interface, data flow, Hadoop I/O, data integrity, compression, serialization, Avro, file-based data structures	9

Unit 4: MapReduce workflows, unit tests with MRUnit, test data and local tests, anatomy of MapReduce job run, classic Map-reduce, YARN, failures in classic Map-reduce and YARN, job scheduling, shuffle and sort, task execution, MapReduce types, input formats, output formats	10
Unit 5:	7
Hbase, data model and implementations, Hbase clients, Hbase examples, praxis.Cassandra, Cassandradata model, Cassandra examples,Cassandra clients, Hadoop integration.	
Unit 6: Pig, Grunt, pig data model, Pig Latin, developing and testing Pig Latin scripts. Hive, data types and file formats, HiveQL data definition, HiveQL datamanipulation, HiveQL queries.	6

References:

1. Michael Minelli, Michelle Chambers, and AmbigaDhiraj, "Big Data, Big Analytics: Emerging
2. V.K. Jain, Big Data and Hadoop, Khanna Publishing House, New Delhi (2017).
3. V.K. Jain, Data Analysis, Khanna Publishing House, New Delhi (2019).
4. Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013.
5. P. J. Sadalage and M. Fowler, "NoSQL Distilled: A Brief Guide to the EmergingWorld of Polyglot Persistence", Addison-Wesley Professional, 2012.
6. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012.
7. Eric Sammer, "Hadoop Operations", O'Reilley, 2012.
8. E. Capriolo, D. Wampler, and J. Rutherglen, "Programming Hive", O'Reilley, 2012.
9. Lars George, "HBase: The Definitive Guide", O'Reilley, 2011.
10. Eben Hewitt, "Cassandra: The Definitive Guide", O'Reilley, 2010.
11. Alan Gates, "Programming Pig", O'Reilley, 2011.

Cyber Law and Ethics

Code: OEC-CSBS601B
Contacts: 3L

Name of the Course:	**Cyber Law and Ethics**	
Course Code: OEC-CSBS601B	Semester:VI	
Duration:6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory: 3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz : 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam: 70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction of Cybercrime:** What is cybercrime?, Forgery, Hacking, Software Piracy, Computer Network intrusion[4L]. **Category of Cybercrime:** how criminals plan attacks, passive attack, Active attacks, cyberstalking. [4L]	8	
2	**Cybercrime Mobile & Wireless devices:** Security challenges posted by mobile devices, cryptographic security for mobile devices, Attacks on mobile/cellphones, Theft, Virus, Hacking. Bluetooth; Different viruses on laptop [8L]	8	
3	**Tools and Methods used in Cyber crime:** Proxy servers, panword checking, Random checking, TrojanHorses and Backdoors; DOS & DDOS attacks; SQL injection: buffer over flow. [8L]	8	
4.	**Phishing & Identity Theft:** Phising methods, IDTheft; Online identity method. [4L] **Cybercrime & Cybersecurity:** Legal aspects, indianlaws, IT act, Public key certificate. [4L]	8	

Text book and Reference books:

1. Cyber security by Nina Gobole & Sunit Belapune; Pub: Wiley India.
2. Information Security & Cyber laws, Gupta & Gupta, Khanna Publishing House

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Mobile Computing

Code: OEC-CSBS601C
Contacts: 3L

Name of the Course:	**Mobile Computing**	
Course Code: OEC-CSBS601C	Semester: VI	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: 3L		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam: 70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/Unit
1	Introduction to Personal Communications Services (PCS): PCS Architecture, Mobility management, Networks signalling. Global System for Mobile Communication (GSM) system overview: GSM Architecture, Mobility management, Network signalling.	5	
2	General Packet Radio Services (GPRS): GPRS Architecture, GPRS Network Nodes. Mobile Data Communication: WLANs (Wireless LANs) IEEE 802.11 standard, Mobile IP.	5	
3	Wireless Application Protocol (WAP): The Mobile Internet standard, WAP Gateway and Protocols, wireless mark up Languages (WML). Wireless Local Loop(WLL): Introduction to WLL Architecture, wireless Local Loop Technologies.	7	
4.	Third Generation (3G) Mobile Services: Introduction to International Mobile Telecommunications 2000 (IMT 2000) vision, Wideband Code Division Multiple Access (W-CDMA), and CDMA 2000, Quality of services in 3G	7	
5	Global Mobile Satellite Systems; case studies of theIRIDIUM and GLOBALSTAR systems. Wireless Enterprise Networks: Introduction to Virtual Networks,Blue tooth technology, Blue tooth Protocols.	7	
6	Server-side programming in Java, Pervasive web application architecture, Device independent example	8	

	application		

Text book and Reference books:

1. "Pervasive Computing", Burkhardt, Pearson
2. "Mobile Communication", J. Schiller, Pearson
3. "Wireless and Mobile Networks Architectures", Yi-Bing Lin & Imrich Chlamtac, John Wiley& Sons, 2001
4. "Mobile and Personal Communication systems and services", Raj Pandya, Prentice Hall ofIndia, 2001.
5. "Guide to Designing and Implementing wireless LANs", Mark Ciampa, Thomson learning,Vikas Publishing House, 2001.
6. "Wireless Web Development", Ray Rischpater, Springer Publishing,
7. "The Wireless Application Protocol", Sandeep Singhal, Pearson .
8. "Third Generation Mobile Telecommunication systems", by P.Stavronlakis, Springer Publishers,
9. Brijesh Gupta “Mobile Computing”, Khanna Publishing House, New Delhi

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Bio Informatics
Code: OEC-CSBS601D
Contacts: 3L

Name of the Course:	**Bio Informatics**	
Course Code: OEC-CSBS601D	Semester: VI	
Duration: 6 months	Maximum Marks: 100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam: 70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/Unit
1	**INTRODUCTION TO MOLECULAR BIOLOGY** Concepts of Cell, tissue, types of cell, components of cell, organelle. Functions of different organelles. Concepts of DNA: Basic Structure of DNA; Double Helix structure; Watson and crick model. Exons and Introns and Gene Concept. Concepts of RNA : Basic structure, Difference between RNA and DNA. Typesof RNA. Concept of Protein: Basic components and structure. Introduction to Central Dogma: Transcription and Tranlation Introduction to Metabolic Pathways.	5	
2	**Sequence Databases** Introduction to Bioinformatics. Recent challenges in Bioinformatics. Protein Sequence Databases, DNA sequence databases. sequence database search programs like BLAST and FASTA. NCBI different modules: GenBank; OMIM, Taxonomybrowser, PubMed;	2	
3	**DNA SEQUENCE ANALYSIS** DNA Mapping and Assembly : Size of Human DNA ,Copying DNA: Polymerase Chain Reaction (PCR), Hybridization and Microarrays, Cutting DNA into Fragments, Sequencing Short DNA Molecules, Mapping Long DNA Molecules. DeBruijn Graph. Sequence Alignment: Introduction, local and global alignment, pair wise and multiple alignment, Dynamic Programming Concept. Alignment algorithms: Needleman and Wunsch algorithm, Smith-Waterman.	14	

4.	**Introduction Probabilistic models used in Computational Biology** Probabilistic Models; Hidden Markov Model : Concepts, Architecture, Transition matrix, estimation matrix. Application of HMM in Bioinformatics : Genefinding, profile searches, multiple sequence alignment and regulatory site identification. Bayesian networks Model :Architecture, Principle ,Application in Bioinformatics.	8	
5.	**Biological Data Classification and Clustering** Assigning protein function and predicting splice sites:Decision Tree	6	

Numerical Methods
Code: OEC-CSBS601E
Contact: 3L

Name of the Course:	**Numerical Methods**	
Course Code: OEC-CSBS601E	Semester: VI	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam:70 Marks
Credit Points:	3	

Unit	Content	Hrs/Unit	Marks/Unit
1	Approximation in numerical computation: Truncation and rounding errors, Fixed and floating-point arithmetic, Propagation of errors.	2	
2	Interpolation: Newton forward/backward interpolation, Lagrange's and Newton's divideddifference Interpolation.	8	
3	Numerical integration: Trapezoidal rule, Simpson's1/3 rule, Expression for corresponding error terms.	3	

4.	Numerical solution of a system of linear equations: Gauss elimination method, Matrix inversion, LU Factorization method, Gauss-Seidel iterative method.	8	
5	Numerical solution of Algebraic equation: Bisection method, Regula-Falsi method,Newton-Raphson method.	3	
6	Numerical solution of ordinary differential equation:Euler's method, Runge-Kutta methods, Predictor-Corrector methods and Finite Difference method.	2	

Text book and Reference books:

1. R.S. Salaria: Computer Oriented Numerical Methods, Khanna Publishing House
2. C.Xavier: C Language and Numerical Methods.
3. Dutta & Jana: Introductory Numerical Analysis.
4. J.B.Scarborough: Numerical Mathematical Analysis.
5. Jain, Iyengar , & Jain: Numerical Methods (Problems and Solution).
6. Balagurusamy: Numerical Methods, Scitech.
7. Baburam: Numerical Methods, Pearson Education.
8. N. Dutta: Computer Programming & Numerical Analysis, Universities Press.

Introduction to Industrial Management (Humanities III)

Code: HSMC-CSBS601

Contacts: 2L

Name of the Course:	**Introduction to Industrial Management (Humanities III)**	
Course Code: HSMC-CSBS601	Semester: VI	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:2 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz : 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam:70 Marks
Credit Points:	2	

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Unit	Content	Hrs/Unit	Marks/Unit
1	**Introduction** System- concept, definition, types, parameters, variables and behavior. Management – definition andfunctions. Organization structure: i. Definition. ii. Goals. iii. Factors considered in formulatingstructure. iv. Types. v. Advantages and disadvantages. vi. Applications. Concept, meaning and importance ofdivision of labor, scalar & functional processes, span of control, delegation ofauthority, centralization and decentralization in industrial management. Organizational culture and climate –meaning, differences and factors affecting them. Moral-factors affecting moral. Relationship between moral andproductivity. Job satisfaction- factors influencingjob satisfaction. Important provisions of factory actand labor laws.	6	
2	**Critical Path Method (CPM) andProgramme Evaluation Review Technique (PERT):** 2.1 CPM & PERT-meaning, features, difference, applications. 2.2 Understand different terms used in network diagram. Draw network diagram for a real lifeproject containing 10-15 activities, computation of LPO and EPO.(Take minimum three examples). Determination of critical path onnetwork. Floats, its types and determination offloats. Crashing of network, updating andits applications.	6	

3	**Materials Management:** Material management-definition, functions, importance, relationship withother departments. Purchase - objectives, purchasing systems, purchase procedure, terms andforms used in purchase department. Storekeeping- functions, classification of stores as centralized and decentralized with their advantages, disadvantages and application in actual practice. Functions of store, types of records maintained by store, various types and applications of storage equipment, needand general methods for codification ofstores. Inventory control: i. Definition. ii. Objectives. iii. Derivation for expression for Economic Order Quantity (EOQ) and numeric examples. iv. ABC analysis andother modern methods of analysis. v. Various types of inventory models such as Wilson's inventory model, replenishment model and two bin model.(Only sketch and understanding, no derivation.). 3.6 Material Requirement Planning (MRP)- concept, applications and brief details about software packages availablein market.	6	

4	**Production planning and Control(PPC):** Types and examples of production.PPC : i. Need and importance. ii. Functions. iii. Forms used and their importance. iv. General approach foreach type of production. Scheduling- meaning and need forproductivity and utilisation. Gantt chart- Format and method toprepare. Critical ratio scheduling-method andnumeric examples. Scheduling using Gantt Chart (for atleast 5-7 components having 5-6	6	
	machining operations, with processes, setting and operation time for each component and process, resources available, quantity and other necessarydata), At least two examples. 4.7 Bottlenecking- meaning, effect andways to reduce.		
5	**Value Analysis (VA) and Cost Control:** 5.1 VA-definition, terms used, process andimportance. 5.2 VA flow diagram. DARSIRI method of VA. Case study of VA-at least two. Waste-types, sources and ways to reduce them. Cost control-methods and important guide lines.	3	
6	**Recent Trends in IM:** ERP (Enterprise resource planning) - concept,features and applications. Important features of MS Project. Logistics- concept, need and benefits. Just in Time (JIT)-concept and benefits. Supply chain management-concept and benefits.	3	

Text book and Reference books:

1. L.S. Srinath– "CPM & PERT principles and Applications".
2. Buffa – "Modern Production Management".
3. N. Nair – "Materials Management".
4. O. P. Khanna – " Industrial Engineering & Management".
5. Mikes – "Value Analysis".
6. S.C. Sharma, "Engineering Management – Industrial Engineering & Management",Khanna Book Publishing Company, New Delhi

Course Outcomes:

On completion of the course students will be able to

1. Interpret given organization structure, culture, climate and major provisions of factoryacts and laws.
2. Explain material requirement planning and store keeping procedure.
3. Plot and analyze inventory control models and techniques.
4. Prepare and analyze CPM and PERT for given activities.
5. List and explain PPC functions.

Design and Analysis of Algorithms Lab

Code: PCC-CSBS691
Contact: 4P

Name of the Course:	**Design and Analysis of Algorithms Lab**
Course Code: PCC-CSBS691	Semester: VI
Duration:6 months	Maximum Marks:100
Teaching Scheme:	
Theory: hrs./week	Continuous Internal Assessment
Tutorial: NIL	External Assessment: 60
Practical: 4 hrs./week	Distribution of marks: 40
Credit Points:	2
Pre-Requisite:	
Pre-Requisite as in : PCC-CSBS601	

Laboratory Experiments:	
Divide and Conquer:	
1	Implement Binary Search using Divide and Conquer approach Implement Merge Sort using Divide and Conquer approach
2	Implement Quick Sort using Divide and Conquer approach Find Maximum and Minimum element from a array of integer using Divide and Conquer approach
3	Find the minimum number of scalar multiplication needed for chain of matrix

4	Implement all pair of Shortest path for a graph (Floyed- Warshall Algorithm) Implement Traveling Salesman Problem
5	Implement Single Source shortest Path for a graph (Dijkstra , Bellman Ford Algorithm
Brunch and Bound:	
6	Implement 15 Puzzle Problem
Backtracking:	
7	Implement 8 Queen problem
8	Graph Coloring Problem Hamiltonian Problem
Greedy method	
9	Knapsack Problem Job sequencing with deadlines
10	Minimum Cost Spanning Tree by Prim's Algorithm Minimum Cost Spanning Tree by Kruskal's Algorithm
Graph Traversal Algorithm:	
11	Implement Breadth First Search (BFS)
	Implement Depth First Search (DFS)

Any experiment specially designed by the college
(Detailed instructions for Laboratory Manual to be followed for further guidance)

Operating System Lab
Code: PCC-CSBS692
Contacts: 4P

Name of the Course:	Operating System Lab
Course Code: PCC-CSBS692	Semester: VI
Duration:6 months	Maximum Marks:100
Teaching Scheme:	
Theory: hrs./week	Continuous Internal Assessment
Tutorial: NIL	External Assesement:60
Practical: 4 hrs./week	Distribution of marks:40
Credit Points:	2

1 1. **Managing Unix/Linux Operating System [8P]:**
Creating a bash shell script, making a script executable, shell syntax (variables, conditions,control structures, functions,

commands). Partitions, Swap space, Device files, Raw and Block files, Formatting disks,Making file systems, Superblock, I-nodes, File system checker, Mounting file systems,
Logical Volumes, Network File systems, Backup schedules and
methods Kernel loading, init and the inittab file, Run-levels, Run level scripts. Passwordfile management, Password
security, Shadow file, Groups and the group file, Shells, restricted shells, user-managementcommands, homes and
permissions, default files, profiles, locking accounts, setting passwords, Switching user,Switching group, Removing users &user groups.

2. **Process [4P]**: starting new process, replacing a process image, duplicating a processimage, waiting for a process,
zombie process.
3. **Signal [4P]**: signal handling, sending signals, signal interface, signal sets.
4. **Semaphore [6P]**: programming with semaphores (use functions semctl, semget, semop,set_semvalue, del_semvalue, semaphore_p, semaphore_v).
5. **POSIX Threads [6P]**: programming with pthread functions (viz. pthread_create,pthread_join, pthread_exit,
pthread_attr_init, pthread_cancel)
6. **Inter-process communication [6P]**: pipes(use functions pipe, popen, pclose), namedpipes(FIFOs, accessing FIFO),
message passing & shared memory(IPC version V).

Any experiment specially designed by the college
(Detailed instructions for Laboratory Manual to be followed for further guidance)

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Semester-VII

Introduction to IoT
Code: PEC-CSBS701A
Credit: 3

Course Outcome(s):

This course will help students understand basic principles and concepts of Internet-of-Things use cases, applications, architecture and technologies. Students will get an overview of an end to end IoT system encompassing the edge, cloud and application tiers. This course will build upon the foundations created in the pre-requisite courses and will equip the students to architect a complete IoT application on their own. The lab exercises will consist of hands-on experiments that will lead to building an IoT application end-to-end. Some of the specialized topics will be covered via student seminars where students are expected to research and present their findings in a seminar format.

Topics to Be Covered:

UNIT – I

Introduction to IoT and Use cases: Understanding basic concepts of IoT, Consumer IoT vs Industrial Internet, Fundamental building blocks, Use Cases of IoT in various industry domains,

UNIT – II

Architecture: IoT reference architectures, Industrial Internet Reference Architecture, Edge Computing, IoT Gateways, Data Ingestion and Data Processing Pipelines, Data Stream Processing

UNIT – III

Sensors and Industrial Systems: Introduction to sensors and transducers, integrating sensors to sensor processing boards, introduction to industrial data acquisition systems, industrial control systems and their functions

UNIT – IV

Networking and Communication for IoT: Recap of OSI 7 layer architecture and mapping to IoT architecture, Introduction to proximity networking technologies (ZigBee, Bluetooth, Serial Communication), Industrial network protocols (Modbus, CANbus), Communicating with cloud applications (web services, REST, TCP/IP and UDP/IP sockets, MQTT, WebSockets, protocols. Message encoding (JSON, Protocol Buffers)

UNIT – V

IoT Data Processing and Storage: Time Series Data and their characteristics, time series databases, basic time series analytics, data summarization and sketching, dealing with noisy and missing data, anomaly and outlier detection,

IoT Seminars:

Selected topics in IoT should be handled via student seminars. Recommended that students form a group do research on at least one of the following topics and present it through seminars. They are expected to do a literature survey of the topic and present their survey paper to the class. The suggested topics are –

a) IoT Applications

- Smart Cities
- Connected Vehicles and Telematics
- Smart Grids
- Smart Homes

b) IoT data visualization

c) Survey of cloud based IoT platforms

d) Low power wide area networks for IoT

e) IoT device management

f) Survey of chips, embedded modules and development boards for IoT devices

g) Embedded and real-time operating systems for IoT

h) IoT Security

- Security risks in IoT
- Securing IoT endpoint devices and secure communication protocols for IoT
- Security and Privacy of IoT data

HUMAN RESOURCE MANAGEMENT
Code: HSMC-CSBS702
Credit: 2

Course Outcomes On completion of this course, the students will be able

CO1: To develop the understanding of the concept of human resource management and to understand its relevance in organizations.
CO2: To develop necessary skill set for application of various HR issues.
CO3: To analyse the strategic issues and strategies required to select and develop manpower resources.
CO4: To integrate the knowledge of HR concepts to take correct business decisions.

MODULE I

1. Human Resource Management: Meaning, Scope, objectives, and functions of HRM , HR as a Factor of Competitive Advantage, Structure of HR Department, , Line and staff responsibility of HR Managers, Environmental factors influencing HRM (2L)

2. Human Resource Planning: definition, objective, process of HRP. Supply and Demand Forecasting techniques, Manpower Inventory, Career Planning& Development, Succession Planning, Rightsizing, Restructuring. Human Resource Information System (HRIS) (6L)

3. Recruitment and Selection: Process, Sources, Methods of selection, Interviewing Methods, Skills and Errors. (4L)

4. Human Resource Development: Definition, objective, process of HRD, Assessment of HRD Needs, HRD Methods: Training and Non-Training, Training Process; Designing, Implementation and Evaluation of Training Programmes, Induction Training. Developing Managerial Skills for: team management, collaboration, interaction across business functions, presentation , Negotiation, and Networking (6L)

5. Performance Appraisal Systems : Purpose, Methods, Appraisal instruments, 360 degree Appraisal, HR Score Card, Errors in appraisal, Potential Appraisal, Appraisal Interview. (4L)

MODULE –II

6. Compensation Management : Concepts, Components; System of Wage Payment, job evaluation, wage/ salary fixation, incentives, bonus, ESOPs, Fringe Benefits, Retirement Benefits. Compensation Plans (4L)

7. Industrial Relations in India: Parties; Management and Trade Unions, Industrial Disputes: Trends, Collective Bargaining, Settlement Machineries, Role of Government, Labour Policy in India. (4L)

8. Workers' Participation in Management: Concept, Practices and Prospects in India, Quality Circles and other Small Group Activities. (2L)

9. Discipline Management: Misconduct, Disciplinary action, Domestic Enquiry, Grievance Handling (4L)

10. Strategic HRM: Meaning, Strategic HRM vs Traditional HRM, SHRM Process, barriers to SHRM. Nature of e-HRM, e-Recruitment & Selection, e-Performance Management, e-Learning (4L)

Suggested Readings:

1. Agarwala T. - Strategic Human Resource Management, OUP
2. Aswathappa, K. - Human Resource Management, Tata McGraw Hill
3. Jyothi P. & Venkatesh, D.N. - Human Resource Management, OUP
4. Ramaswamy, E.A. - Managing Human Resources, OUP
5. Saiyadain, M.S - Human Resource Management : Tata McGraw Hill
6. Mondal Sabari & Goswami Amal - Human Resource Management: Vrinda Publications

FINANCIAL & COST ACCOUNTING
Code: HSMC-CSBS701
Credit: 2

Course Outcome(s):

This course will help students

• To create an awareness about the importance and usefulness of the accounting concepts and their managerial implications

• To develop an understanding of the financial statements and the underlying principles and learn to interpret financial statements

• To create awareness about cost accounting, different types of costing and cost management

Topics to Be Covered:

UNIT – I Accounting Concept: Introduction, Techniques and Conventions, Financial Statements- Understanding & Interpreting Financial Statements

UNIT – II Accounting Process: • Book Keeping and Record Maintenance • Fundamental Principles and Double Entry • Journal, Ledger, Trial Balance, Balance Sheet, Final Accounts • Cash Book and Subsidiary Books • Rectification of Errors

UNIT – III Financial Statements: Form and Contents of Financial Statements, Analyzing and Interpreting Financial Statements, Accounting Standards. Class Discussion: Corporate Accounting Fraud- A Case Study of Satyam

UNIT – IV Cash Flow and Fund Flow Techniques: Introduction, How to prepare, Difference between them

Programming in Matlab
Code: PCC-CSBS791
Credit: 2

Introduction

Why Matlab?, History, Its strengths, Competitors, Starting MATLAB, Using MATLAB as a calculator, Quitting MATLAB

Basics

Familiar with MATLAB windows, Basic Operations, MATLAB-Data types, Rules about variable names, Predefined variables

Programming-I

Vector, Matrix, Array Addressing, Built-in functions, Mathematical Operations, Dealing with strings (Array of characters), Array of array (cell) concept

Programming-II

Script file, Input commands, Output commands, Structure of function file, Inline functions, Feval command, Comparison between script file and function file

Conditional statements and Loop

Relational and Logical Operators, If-else statements, Switch-case statements, For loop, While loop, Special commands (Break and continue), Import data from large database, Export data to own file or database

2D Plotting

In-built functions for plotting, Multiple plotting with special graphics, Curve fitting, Interpolation, Basic fitting interface

3D Plotting

Use of meshgrid function, Mesh plot, Surface plot, Plots with special graphics

Proposed Syllabus for - ADVANCED SOCIAL, TEXT AND MEDIA ANALYTICS

Code: PEC-CSBS701C Credit: 3

Name of the Course:	**Advanced Social, Text and Media Analytics**
Course Code: PEC-CSBS701C	Semester: VII
Duration: 6 months	Maximum Marks: 100
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs./week	Mid Semester exam: 15
Tutorial: NIL Practical: NIL Credit Points: 3	Assignment and Quiz: 10 marks Attendance: 5 marks End Semester Exam: 70 Marks

Course Outcome(s):

CO1: To understand the fundamental concepts and techniques of text mining, text analytics, and natural language processing for extracting useful information from unstructured text.

CO2: To apply web analytics and information retrieval techniques for analysing web data, user behaviour, search processes, and online content.

CO3: To understand social media data, social networks, graph-based representations, and measures used for social network analysis.

CO4: To apply statistical, machine learning, sentiment analysis, topic modelling, and predictive techniques to social, text, and media data.

CO5: To analyse multimedia and digital media data and communicate analytical findings through appropriate visualization and real-world applications.

Module / Unit Content:

UNIT – I	**Text Mining**
	Introduction to text mining and text analytics, sources and characteristics of textual data, core text mining operations, text preprocessing techniques, tokenization, stop-word removal, stemming and lemmatization, n-gram representation, Bag-of-Words model, TF-IDF, text categorization, text clustering, information extraction, probabilistic models for information extraction, text mining applications.

UNIT – II	**Text Analytics and Natural Language Processing**
	Content analysis, Natural Language Processing techniques, linguistic processing, part-of-speech tagging, named entity recognition, document similarity, text classification, clustering and topic detection, topic modelling, predictive modelling for text, sentiment analysis, opinion mining, sentiment prediction and applications of text analytics.

UNIT – III	**Web Analytics and Web Mining**
	Introduction to web analytics, web analytics tools, web data collection, clickstream analysis, A/B testing, online surveys, web search and retrieval, search engine optimization, web crawling and indexing, ranking algorithms, web traffic models, user behaviour analysis, conversion analysis and applications of web analytics.

UNIT – IV	**Social Media Analytics and Social Network Analysis**

	Social media data and methods, social network and web data, network representation using graphs and matrices, nodes, edges and ties, basic measures for individuals and networks, degree, centrality, density and clustering, link analysis, random graphs and network evolution, social contexts, affiliation and identity, community detection, influence and information diffusion, social network analysis of platforms such as X/Twitter, Facebook, YouTube, LinkedIn and Instagram.

UNIT – V	**Media Analytics, Visualization and Applications**
	Introduction to media analytics, digital and multimedia data, image and video analytics fundamentals, feature extraction and classification concepts, media content analysis, social media metrics and engagement analysis, information visualization, visual analytics, dashboards and storytelling with data, campaign and brand analytics, recommendation and personalization, ethical, privacy and security issues in social and media analytics, and case studies in business, marketing, public services and smart applications.

Suggested Readings:

1. Matthew Ganis and Avinash Kohirkar, Social Media Analytics: Techniques and Insights for Extracting Business Value Out of Social Media, Pearson.

2. Ronen Feldman and James Sanger, The Text Mining Handbook: Advanced Approaches in Analyzing Unstructured Data, Cambridge University Press.

3. Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Introduction to Information Retrieval, Cambridge University Press.

4. Charu C. Aggarwal and ChengXiang Zhai (Eds.), Mining Text Data, Springer.

5. Matthew A. Russell, Mining the Social Web, O'Reilly Media.

6. Reza Zafarani, Mohammad Ali Abbasi and Huan Liu, Social Media Mining: An Introduction, Cambridge University Press.

7. Avinash Kaushik, Web Analytics 2.0: The Art of Online Accountability and Science of Customer Centricity, Sybex.

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Semester-VIII

Name of the Course:	**CRYPTOLOGY**	
Course Code: PEC-CSBS801A	Semester: VIII	
Duration:6 months	Maximum Marks:100	
Teaching Scheme		**Examination Scheme**
Theory:3 hrs./week		Mid Semester exam: 15
Tutorial: NIL		Assignment and Quiz: 10 marks
		Attendance: 5 marks
Practical: NIL		End Semester Exam:70 Marks
Credit Points:	3	

UNIT I
INTRODUCTION

Security trends – Legal, Ethical and Professional Aspects of Security, Need for Security at Multiple levels, Security Policies – Model of network security – Security attacks, services and mechanisms – OSI security architecture – Classical encryption techniques: substitution techniques, transposition techniques, steganography- Foundations of modern cryptography: perfect security – information theory – product cryptosystem – cryptanalysis.

UNIT II
SYMMETRIC KEY CRYPTOGRAPHY

MATHEMATICS OF SYMMETRIC KEY CRYPTOGRAPHY: Algebraic structures – Modular arithmetic-Euclid's algorithm- Congruence and matrices -Groups, Rings, Fields- Finite fields- SYMMETRIC KEY CIPHERS: SDES – Block cipher Principles of DES – Strength of DES – Differential and linear cryptanalysis – Block cipher design principles – Block cipher mode of operation – Evaluation criteria for AES – Advanced Encryption Standard – RC4 –
Key distribution.

UNIT III
PUBLIC KEY CRYPTOGRAPHY

MATHEMATICS OF ASYMMETRIC KEY CRYPTOGRAPHY: Primes – Primality Testing –Factorization – Euler's totient function, Fermat's and Euler's Theorem – Chinese Remainder Theorem – Exponentiation and logarithm – ASYMMETRIC KEY CIPHERS: RSA cryptosystem – Key distribution – Key management – Diffie Hellman key exchange -ElGamal cryptosystem – Elliptic curve arithmetic-Elliptic curve cryptography.

UNIT IV
MESSAGE AUTHENTICATION AND INTEGRITY

Authentication requirement – Authentication function – MAC – Hash function – Security of hash function and MAC – SHA –Digital signature

and authentication protocols – DSS- Entity Authentication: Biometrics, Passwords, Challenge Response protocols- Authentication applications – Kerberos, X.509

UNIT V

SECURITY PRACTICE AND SYSTEM SECURITY

Electronic Mail security – PGP, S/MIME – IP security – Web Security – SYSTEM SECURITY: Intruders – Malicious software – viruses – Firewalls.

Name of the Course:		**Quantum Computing**	
Course Code: PEC-CSBS801B		Semester: VIII	
Duration: 6 months		Maximum Marks:100	
Teaching Scheme			**Examination Scheme**
Theory:3 hrs./week			Mid Semester exam: 15
Tutorial: NIL			Assignment and Quiz: 10 marks
			Attendance : 5 marks
Practical: NIL			End Semester Exam :70 Marks
Credit Points:		3	
Objective:			
1	The course will provide an insight of basic of quantum physics from a computer scientist's perspective, and how it describes reality and understand the philosophical implications of quantum computing		
Pre-Requisite:			
1	Linear Algebra, Theory of Computation		

Unit	Content	Hrs/Unit	Marks/ Unit
1	Qubit & Quantum States: The Qubit, Vector Spaces. Linear Combination Of Vectors, Uniqueness of a spanning set, basis & dimensions, inner Products, orthonormality, gram-schmidt orthogonalization, bra-ket formalism, the Cauchyschwarez and triangle Inequalities.	3	
2	Matrices & Operators: Observables, The Pauli Operators, Outer Products, The Closure Relation, Representation of operators using matrices, outer products & matrix representation, matrix representation of operators in two dimensional spaces, Pauli Matrix, Hermitian unitary and normal operator, Eigen values & Eigen Vectors, Spectral Decomposition, Trace of an operator, important properties of Trace, Expectation Value of Operator, Projection Operator, Positive Operators,	10	
3.	Commutator Algebra, Heisenberg uncertainty principle, polar decomposition &singular values, Postulates of Quantum	5	

	Mechanics.		
4.	Tensor Products: Representing Composite States in Quantum Mechanics, Computing inner products, Tensor products of column vectors, operators and tensor products of Matrices. Density Operator: Density Operator of Pure & Mix state, Key Properties, Characterizing Mixed State, Practical Trace & Reduce Density Operator, Density Operator & Bloch Vector.	5	
5.	Quantum Measurement Theory: Distinguishing Quantum states & Measures, Projective Measurements, Measurement on Composite systems, Generalized Measurements, Positive Operator- Valued Measures.	8	
6.	Recent trends in Quantum Computing Research, Quantum Computing Applications of Genetic Programming.	6	

Text book and Reference books:

Quantum Computing without Magic by Zdzislaw Meglicki
2. Quantum Computing Explained By DAVID Mc MAHON
3. Quantum Computer Science By Marco Lanzagorta, Jeffrey Uhlmann
4. An Introduction to Quantum Computing Phillip Kaye, Raymond Laflamme, Michele Mosca.

Course Outcomes:

On completion of the course students will be able to
knowledge of Vector spaces, Matrices, Quantum state, Density operator and Quantum

Services Science & Service Ops Management(HSMCCSBS801)

Course Pre Requisite(s):

Fundamentals of Management, Operations Research

Course Outcome(s):

Students will be able to

- Understand concepts about Services and distinguish it from Goods
- Able to identify characteristics and nature of Services
- Comprehend ways to design Services and evaluate them using Service qualities
- Understand how various methods can be used to operate and manage Service businesses
- Understand how innovation can be approached from Services point of view

Topics to Be Covered:

UNIT –I

Introduction: Introduction to the course, Introduction to service operations, Role of service in economy and society, Introduction to Indian service sector

Nature of Services and Service Encounters: Differences between services and operations, Service package, characteristics, various frameworks to design service operation system, Kind of service encounter, importance of encounters

Service-Dominant Logic: From Goods-Dominant logic to Service-Dominant logic, Value Co-creation

UNIT – II

Service Strategy and Competitiveness: Development of Strategic Service Vision (SSV), Data Envelopment Analysis

New Service Development: NSD cycle, Service Blueprinting, Elements of service delivery system

Service Design: Customer Journey and Service Design, Design Thinking methods to aid Service Design

Locating facilities and designing their layout: models of facility locations (Huff's retail model), Role of service-scape in layout design

Service Quality: SERVQUAL, Walk through Audit, Dimensions of Service quality & other quality tools

UNIT – III

Service Guarantee & Service Recovery: How to provide Service guarantee? How to recover from Service failure?

UNIT – IV

Forecasting Demand for Services: A review of different types of forecasting methods for demand forecasting.

Managing Capacity and Demand: Strategies for matching capacity and demand, Psychology of waiting, Application of various tools used in managing waiting line in services.

Managing Facilitating Goods: Review of inventory models, Role of inventory in services

Managing service supply relationship: Understanding the supply chain/hub of service, Strategies for managing suppliers of service

Vehicle Routing Problem: Managing after sales service, Understanding services that involve transportation of people and vehicle, Techniques for optimizing vehicle routes

UNIT – V

Service Innovation: Services Productivity, Need for Services Innovation

Student Project:

Option 1: Choose any service organization around and present it from the perspective of: nature of service, classification of service, blueprint or service design analysis, service quality, and any additional perspective you would like to add.

Option 2: Choose any latest research paper in services and explain your understanding and feedback on the same.

Text Books:

1. Fitzsimmons & Fitzsimmons, Service Management: Operations, Strategy, Information Technology, McGraw Hill publications (7th edition)

Reference Books:

1. Wilson, A., Zeithaml, V. A., Bitner, M. J., & Gremler, D. D. (2012). *Services marketing: Integrating customer focus across the firm*. McGraw Hill.

2. Lovelock, C. (2011). *Services Marketing, 7/e*. Pearson Education India

3. Reason, Ben, and Lovlie, Lavrans, (2016) Service Design for Business: A Practical Guide to Optimizing the Customer Experience, Pan Macmillan India,

4. Chesbrough, H. (2010). Open services innovation: Rethinking your business to grow and compete in a new era. John Wiley & Sons.

Reference Papers:

1. Karmarkar, U. (2004). Will you survive the services revolution? Harvard Business Review, 100-107.

2. Vargo, S. L., & Lusch, R. F. (2008). From goods to service (s): Divergences and convergences of logics. Industrial marketing management, 37(3), 254-259.

3. Vargo, S. L., & Lusch, R. F. (2008). "Service-Dominant Logic: Continuing the Evolution," Journal of the Academy of Marketing Science (36:1), pp. 1-10

4. Silvestro, R., Fitzgerald, L., Johnston, R., & Voss, C. (1992). Towards a classification of service processes. International journal of service industry management, 3(3), 62-75.

5. Vargo, S. L., Maglio, P. P., & Akaka, M. A. (2008). On value and value co-creation: A service systems and service logic perspective. European management journal, 26(3), 145-152.

6. Shostack, G.L., (1984), "Designing Services That Deliver," Harvard Business Review, January-February 1984, pp. 132-139

7. Evenson, S., & Dubberly, H. (2010). Designing for service: Creating an experience advantage. Introduction to service engineering, 403-413.

8. Edvardsson, B., & Olsson, J. (1996). Key concepts for new service development. Service Industries Journal, 16(2), 140-164.

9. Goldstein, S. M., Johnston, R., Duffy, J., & Rao, J. (2002). The service concept: the missing link in service design research? Journal of Operations management, 20(2), 121-134.

10. Kumar, A., Zope, N. R., & Lokku, D. S. (2014, April). An approach for services design by understanding value requirements, identifying value carriers, developing value proposition, and subsequently realizing value. In Global Conference (SRII), 2014 Annual SRII (pp. 298-304). IEEE.

11. Parasuraman, A., Zeithaml, V.A., and Berry, L.L., (1985), "A Conceptual Model of Service Quality and Its Implications for Future Research," The Journal of marketing, Vol. 49, No. 4, pp. 41-50

12. Cronin, J.J., and Taylor, S.A., (1992), "Measuring Service Quality: A Reexamination and Extension," The Journal of Marketing, Vol. 56, No. 3, pp. 55-68

13. Van Ree, H. J., (2009), Service Quality Indicators for Business Support Services, Ph.D. Thesis, University College London, London.

14. Zope, N. R., Anand, K., & Lokku, D. S. (2014, April). Reviewing Service Quality for IT Services Offerings: Observations in the Light of Service Quality Models & Determinants. In Global Conference (SRII), 2014 Annual SRII (pp. 43-49). IEEE.

15. Heskett, J.L., Jones, T.O., Loveman, G.W., Sasser, W.E., and Schlesinger, L.A., (2008), "Putting the Service-Profit Chain to Work," Best of HBR, Harvard Business Review, July-August 2008, pp. 118-128

16. Clatworthy, S. (2011). Service innovation through touch-points: Development of an innovation toolkit for the first stages of new service development. International Journal of Design, 5(2).

17. Barras, R. (1986). "Towards a Theory of Innovation in Services," Research Policy (15), pp. 161-173.

18. Gustafsson, A., and Johnson, M. (2003). Competing in a Service Economy: How to Create a Competitive Advantage Through Service Development and Innovation, San Francisco: Jossey-Bass.

19. Barrett, M., Davidson, E., Prabhu, J., & Vargo, S. L. (2015). "Service innovation in the digital age: key contributions and future directions". Mis Quarterly, 39(1), 135-154.

20. Lusch, R. F., and Nambisan, S. (2015). "Service Innovation; A Service-Dominant Logic Perspective," MIS Quarterly (39:1), pp.155-175

RESEARCH METHODOLOGIES(PROJCSBS801)

Pre-requisites: NA

COURSE OBJECTIVE(S)

- To understand some basic concepts of research and its methodologies
- To identify appropriate research topics
- To select and define appropriate research problem and parameters
- To prepare a project proposal (to undertake a project)
- To organize and conduct research (advanced project) in a more appropriate manner
- To write a research report and thesis
- To write a research proposal (grants)

Unit 1: (9 hrs)

RESEARCH FORMULATION AND DESIGN

Motivation and objectives – Research methods vs. Methodology. Types of research – Descriptive vs. Analytical, Applied vs. Fundamental, Quantitative vs. Qualitative, Conceptual vs. Empirical, concept of applied and basic research process, criteria of good research. Defining and formulating the research problem, selecting the problem, necessity of defining the problem, importance of literature review in defining a problem, literature review - primary and secondary sources, reviews, monograph, patents, research databases, web as a source, searching the web, critical literature review, identifying gap areas from literature and research database, development of working hypothesis.

Unit 2: (9 hrs)

DATA COLLECTION AND ANALYSIS

Accepts of method validation, observation and collection of data, methods of data collection, sampling methods, data processing and analysis strategies and tools, data analysis, with statically package (Sigma STAT,SPSS for student t-test, ANOVA, etc.), hypothesis testing.

Unit 3: (9 hrs)

RESEARCH ETHICS, IPR AND SCHOLARY PUBLISHING

Ethics - ethical issues, ethical committees (human & animal); IPR - intellectual property rights and patent law, commercialization, copyright, royalty, trade related aspects of intellectual property rights (TRIPS); scholarly publishing - IMRAD concept and design of research paper, citation and acknowledgement, plagiarism, reproducibility and accountability.

Unit 4: (9 hrs)

INTERPRETATION AND REPORT WRITING

Meaning of Interpretation, Technique of Interpretation, Precaution in Interpretation, Significance of Report Writing, Different Steps in Writing Project Report, Layout of the Project/Research Report, Types of Reports, Oral Presentation, Mechanics of Writing a Project/Research Report, Precautions for Writing Research Reports, Conclusions.

RESOURCES:

Text books:

1. Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., 2002. An introduction to ResearchMethodology, RBSA Publishers.
2. Kothari, C.R., 1990. Research Methodology: Methods and Techniques. New Age International.418p.
3. Sinha, S.C. and Dhiman, A.K., 2002. Research Methodology, Ess Ess Publications. 2 volumes.

Reference books and Materials:

1. Trochim, W.M.K., 2005. Research Methods: the concise knowledge base, Atomic Dog Publishing. 270p.
2. Wadehra, B.L. 2000. Law relating to patents, trademarks, copyright designs and geographical indications. Universal Law Publishing.
3. Anthony, M., Graziano, A.M. and Raulin, M.L., 2009. Research Methods: A Process of Inquiry, Allyn and Bacon.
4. Carlos, C.M., 2000. Intellectual property rights, the WTO and developing countries: the TRIPS agreement and policy options. Zed Books, New York.

Business Communication & Value Science

OEC-CSBS801A

Class: (2+0+0)

Credit : 2

COURSE OUTCOMES: Upon completion of the course, the student will be able to

CO1: Recognize the need for life skills and values (Understand)

CO2: Identify own strengths and opportunities (Analyse)

CO3: Recognize the importance of vocabulary enrichment (Understand)

CO4: Understand the basic tenets of communication and apply the basic communication practices in different types of communication (Apply)

CO5: Apply the life skills to different situations (Apply)

CO6: Comprehend the importance of corporate social responsibility(CSR) and recognize the attributes needed to function in a corporate environment (Understand)

CO7: Understand how stress impacts life and work and identify the best practices to manage Stress (Understand)

CO8: Recognize the importance of time management and identify the best practices (Analyse)

Unit 1 Basics of Communication 4

Theories of communication, Verbal and non-Verbal communication, barriers to communication, ways to improve communication.

UNIT 2 LISTENING SKILLS 4

Listening to Product Description-Listening to Inspiration a l speech–Listening Comprehension–Critical Review-Book/Movie Review–Comparative Analysis.

UNIT 3 CRITICALTHINKING 4

Learn Values from Movies–Film on Diversity–Discussion on key take away of the film–Mind Mapping–Conflict Management–Case Study–Time Management.

UNIT 4 HUMANVAL UES AND BUSINESS COMMUNICATION 4

Self-exploration–SWOT Analysis and Goal setting–Values of individuals – Self-work with immersion (Interviewing Community Workers) and narrate what you think are the values that drive them –Self-awareness; Stress management.

UNIT 5 WORK PLACE ENVIRONMENT AND BUSINESS WRITING 4

Emotional Intelligence–Public Speaking at workplace and real life scenarios–Role Play –Form a land Business letters–Building Professional is min Work Place.

UNIT 6 ETHICS & SOCIAL RESPONSIBILITY 4

Importance of Corporate Social Responsibility (CSR)–Professional Ethics–Organizational Structure in Workplace–Developing Leadership Qualities, Fundamental of empathy, attitude and motivation.

UNIT 7 FEEDBACK AND CUSTOMER RELATIONS 4

Best practices to share and receive feedback–Addressing the Grievances of Customer–Practices to Maintain Customer Satisfaction–Customer Survey.

UNIT 8 HR MANAGEMENT 4

Interview Etiquette–Mock Interview–Entrance & Exit Meetings – Drafting Appointment letters – Drafting Joining Letters.

Maulana Abul Kalam Azad University of Technology, West Bengal
(Formerly West Bengal University of Technology)
Syllabus of B. Tech. in Computer Science and Business Systems
(Applicable from the Academic Session 2020-2021)

OEC-CSBS801B: Financial Management

Course Objectives

1. To develop the knowledge of business finance and management decision.
2. To know the different concepts of long term and short term finance and measurement ofcost of capital.
3. To know the definition, assumption and symbols relating to capital structure theories.
4. To know the meaning and importance of capital budgeting decisions and the concept ofcapital budgeting appraisal methods.
5. To know the concept and classification of working capital and importance of workingcapital management.
6. To know the concept and types of dividend and different dividend theories.

Course Outcomes (CO):

SL NO.	Course Outcome	Mapped Modules
1	Ability to determine the value and wealth maximizationof business and scope of financial management.	Module I – Unit 1
2	Abilityto calculate the cost of equity shares, preferenceshare, debenture and other sources of finance.	Module I – Unit 2
3	Ability to determine the relationship between leverageand cost of capital as per major capital structure theories.	Module I – Unit 3
4	Ability to estimate the required return of projects as per different capital budgeting methods and evaluatinginvestment decisions.	Module II – Unit 4
5	Ability to compute working capital using both the cashcost approach and the operating cycle approach.	Module II – Unit 5
6	Ability to evaluate the impact of different dividend policy on share price and also the significance of bothrelevance and irrelevance theory.	Module II – Unit 6

MODULE 1

Unit 1: **Introduction to Financial Management** **(8L)**

Meaning, core elements, objectives and scope –Role of Finance Manager –Concept of time value of money –Techniques for dealing with time value of money.

Unit 2: **Sources of Finance** **(10L)**

Cost of Capital: Different sources of finance: Long term and short term sources –Cost of Capital: Meaning, relevance and classification –Calculation of specific cost of capital and weighted cost ofcapital.

Unit 3: **Capital Structure Analysis** **(12L)**

Capital Structure-Concept, importance and determinants, Theories of Capital Structure (Net Income, Net Operating Income, MM Hypothesis, Traditional Approach), Concepts of DOL, DFL, DTL and Trading on equity.

MODULE 2

Unit 4: **Capital Budgeting Decision** **(14L)**

Meaning and importance of capital budgeting decisions –Capital budgeting appraisal methods: Payback period method, accounting rate of return method, net present value method, internal rate of return method and profitability index method –Capital rationing.

Unit 5: **Working Capital Management** **(8L)**

Meaning and classification of working capital –Concept and importance of working capital management –Determinants of working capital requirement –Estimating working capital requirement.

Unit 6: **Dividend Decisions** **(8L)**

Meaning, nature and types of dividend, some dividend policies, Determinants of dividend policy, Dividend theories: Walter's model, Gordon's model, Modigliani and Miller Irrelevance theory.

Suggested Readings:

1. Chandra, P., Financial Management: Theory and Practice, TMH.
2. Khan, M.Y. and Jain, P.K., Financial Management: Text,

Problems and Cases, Tata McGrawHill Publishing Co., Ltd.

3. Pandey, I.M., Financial Management, Vikas Publishing House Pvt. Ltd.
4. Sur, D. and Sarkhel, J., An Introduction to Financial Management, Book Syndicate
5. Parasuraman, Financial Management: A Step by Step Approach, Cengage